we can expand the solution to Eq. (7.120) in terms of them;

$$\psi(\mathbf{r}, \tau) = \sum_m A_m(\tau)\phi_m(\mathbf{r}) .$$

Since $\psi(\mathbf{r}, \tau)$ depends on τ, so will the functions $A_m(\tau)$. Now

$$H\psi = H\left[\sum_m A_m(\tau)\phi_m(\mathbf{r})\right] = \sum_m A_m(\tau)H\phi_m(\mathbf{r})$$
$$= \sum_m \lambda_m A_m(\tau)\phi_m(\mathbf{r}) .$$

Also,

$$\frac{\partial\psi}{\partial\tau} = \sum_m \frac{\partial A_m(\tau)}{\partial\tau}\phi_m(\mathbf{r}) .$$

Thus Eq. (7.120) becomes

$$\sum_m \left[\lambda_m A_m(\tau) + \frac{dA_m(\tau)}{d\tau}\right]\phi_m(\mathbf{r}) = 0 .$$

Since the $\phi_m(\mathbf{r})$ are complete and orthonormal, we conclude immediately that

$$\frac{dA_m(\tau)}{d\tau} + \lambda_m A_m(\tau) = 0$$

for all m. The solution for $A_m(\tau)$ is simply

$$A_m(\tau) = A_m(0)e^{-\lambda_m\tau} ,$$

and thus we obtain

$$\psi(\mathbf{r}, \tau) = \sum_m A_m(0)\phi_m(\mathbf{r})e^{-\lambda_m\tau} .$$

Now, to complete the solution to a partial differential equation which is of first order in the time derivative, we need to specify the boundary condition at $\tau = 0$. Assuming that we know $\psi(\mathbf{r}, 0)$ (for example, in the case of the heat equation, if the initial temperature distribution $T(\mathbf{r}, 0)$ is known), we may write

$$\psi(\mathbf{r}, 0) = \sum_m A_m(0)\phi_m(\mathbf{r}) .$$

Using the fact that the ϕ_m form a complete orthonormal set, we have

$$A_n(0) = \int \phi_n^*(\mathbf{r})\psi(\mathbf{r}, 0)\, d^3\mathbf{r} .$$

Thus

$$\psi(\mathbf{r}, \tau) = \sum_m e^{-\lambda_m\tau}\phi_m(\mathbf{r}) \int \phi_m^*(\mathbf{r}')\psi(\mathbf{r}', 0)\, d^3\mathbf{r}' ,$$

or

$$\psi(\mathbf{r}, \tau) = \int G_1(\mathbf{r}, \mathbf{r}', \tau)\psi(\mathbf{r}', 0)\, d^3\mathbf{r}' , \tag{7.121a}$$

where

$$G_1(\mathbf{r}, \mathbf{r}', \tau) = \sum_m \phi_m(\mathbf{r})\phi_m^*(\mathbf{r}')e^{-\lambda_m \tau} . \tag{7.121b}$$

Note that

$$G_1(\mathbf{r}, \mathbf{r}', 0) = \sum_m \phi_m(\mathbf{r})\phi_m^*(\mathbf{r}') = \delta^3(\mathbf{r} - \mathbf{r}') , \tag{7.122}$$

as it must, of course, because of Eq. (7.121a). Also,

$$\left(H + \frac{\partial}{\partial \tau}\right) G_1(\mathbf{r}, \mathbf{r}', \tau) = 0 . \tag{7.123}$$

Thus $\psi(\mathbf{r}, \tau)$, as given by Eq. (7.121a), is indeed a solution which satisfies the prescribed boundary condition.

We have put a subscript 1 on our function $G_1(\mathbf{r}, \mathbf{r}', \tau)$ to denote that it arises from an equation which is first order in time. We will refer loosely to G_1 as a Green's function, although it does not satisfy

$$\left[H + \frac{\partial}{\partial \tau}\right] G(\mathbf{r}, \mathbf{r}', \tau) = \delta^3(\mathbf{r} - \mathbf{r}')\delta(\tau) . \tag{7.124}$$

We will soon see, however, that G_1 is closely related to a function which *is* a Green's function in this usual sense.

Since G_1 does not satisfy Eq. (7.124), but, rather, Eq. (7.123), G_1 is itself a solution to Eq. (7.120). In fact, if $\psi(\mathbf{r}, 0) = \delta(\mathbf{r} - \mathbf{r}_0)$, then Eq. (7.121a) yields immediately

$$\psi(\mathbf{r}, \tau) = G_1(\mathbf{r}, \mathbf{r}_0, \tau)$$

for $\tau \geq 0$. Thus G_1 is a solution of Eq. (7.120) for a very special initial condition: the so-called "point source" initial condition. This fact leads to a simple physical interpretation of the Green's function which we shall discuss shortly.

Needless to say, our choice of initial time was completely arbitrary. We could equally well have chosen the initial time to be τ', and then we would have

$$\psi(\mathbf{r}, \tau) = \int G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')\psi(\mathbf{r}', \tau')\, d^3\mathbf{r}' , \tag{7.125a}$$

where

$$G_1(\mathbf{r}, \mathbf{r}', \tau, \tau') = \sum_m \phi_m(\mathbf{r})\phi_m^*(\mathbf{r}')e^{-\lambda_m(\tau - \tau')} . \tag{7.125b}$$

In other words, $G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')$ *propagates* the function ψ in time from τ' to $\tau > \tau'$; for this reason, G_1 is often referred to as a *propagator*. This manner of speaking suggests that if we propagate ψ from τ_0 to τ_1 and then from τ_1 to τ_2, we should get the same result as if we propagate directly from τ_0 to τ_2.

Mathematically, we write

$$\psi(\mathbf{r}, \tau_1) = \int G_1(\mathbf{r}, \mathbf{r}'', \tau_1, \tau_0)\psi(\mathbf{r}'', \tau_0)\, d^3\mathbf{r}''$$

$$\psi(\mathbf{r}, \tau_2) = \int G_1(\mathbf{r}, \mathbf{r}', \tau_2, \tau_1)\psi(\mathbf{r}', \tau_1)\, d^3\mathbf{r}'$$

$$= \int\int G_1(\mathbf{r}, \mathbf{r}', \tau_2, \tau_1)G_1(\mathbf{r}', \mathbf{r}'', \tau_1, \tau_0)\psi(\mathbf{r}'', \tau_0)\, d^3\mathbf{r}'\, d^3\mathbf{r}'' .$$

Using Eq. (7.125b), we find that

$$\int G_1(\mathbf{r}, \mathbf{r}', \tau_2, \tau_1)G_1(\mathbf{r}', \mathbf{r}'', \tau_1, \tau_0)\, d^3\mathbf{r}'$$

$$= \int \sum_m \phi_m(\mathbf{r})\phi_m^*(\mathbf{r}')e^{-\lambda_m(\tau_2-\tau_1)} \sum_n \phi_n(\mathbf{r}')\phi_n^*(\mathbf{r}'')e^{-\lambda_n(\tau_1-\tau_0)}\, d^3\mathbf{r}' .$$

Since

$$\int \phi_m^*(\mathbf{r}')\phi_n(\mathbf{r}')\, d^3\mathbf{r}' = \delta_{mn} ,$$

we obtain immediately

$$\int G_1(\mathbf{r}, \mathbf{r}', \tau_2, \tau_1)G_1(\mathbf{r}', \mathbf{r}'', \tau_1, \tau_0)\, d^3\mathbf{r}'$$

$$= \sum_m \phi_m(\mathbf{r})\phi_m^*(\mathbf{r}'')e^{-\lambda_m(\tau_2-\tau_1)}e^{-\lambda_m(\tau_1-\tau_0)}$$

$$= \sum_m \phi_m(\mathbf{r})\phi_m^*(\mathbf{r}'')e^{-\lambda_m(\tau_2-\tau_0)}$$

$$= G_1(\mathbf{r}, \mathbf{r}'', \tau_2, \tau_0) .$$

Thus, as we expected,

$$\psi(\mathbf{r}, \tau_2) = \int G_1(\mathbf{r}, \mathbf{r}'', \tau_2, \tau_0)\psi(\mathbf{r}'', \tau_0)\, d^3\mathbf{r}'' .$$

This idea of compounding propagations is a very important notion in physics; we restate the result below:

$$G_1(\mathbf{r}, \mathbf{r}'', \tau_2, \tau_0) = \int G_1(\mathbf{r}, \mathbf{r}', \tau_2, \tau_1)G_1(\mathbf{r}', \mathbf{r}'', \tau_1, \tau_0)\, d^3\mathbf{r}' .$$

Let us now determine the Green's function for an equation of the diffusion or heat conduction type. We work in one dimension for convenience, in the interval $-L/2 \leq x \leq L/2$, and impose periodic boundary conditions, $\psi(L/2, \tau) = \psi(-L/2, \tau)$. As mentioned above, the operator H is simply $-d^2/dx^2$ in this case, so we want to solve

$$-\frac{d^2\phi_m}{dx^2} = \lambda_m\phi_m(x) .$$

This equation has as its solution

$$\phi_m(x) = \frac{1}{\sqrt{L}} e^{i\sqrt{\lambda_m}\tau},$$

where $\lambda_m = (2\pi m/L)^2$, in order to satisfy the boundary condition, and m is any integer. Thus if we call $k_m = 2\pi m/L$, we get for $G_1(x, x', \tau)$, using Eq. (7.121b),

$$G_1(x, x', \tau) = \sum_{m=-\infty}^{\infty} \frac{1}{L} e^{ik_m(x-x')} e^{-k_m^2\tau}. \tag{7.126}$$

Now suppose that we want to obtain the Green's function on the *infinite* line. Let us employ the same type of limiting procedure used in making the transition from Fourier series to Fourier integrals in Section 5.7. In Eq. (7.126), as L becomes larger and larger, k_m changes only very slightly as we go from one value of m to the next in the sum. In fact, k_m changes by $2\pi/L$ each time m is incremented by one unit. Let us therefore set $2\pi/L = \Delta k_m$ and write

$$G_1(x, x', \tau) = \frac{1}{2\pi} \sum_{m=-\infty}^{\infty} \Delta k_m e^{ik_m(x-x')} e^{-k_m^2\tau}.$$

In this form we see that our expression is exactly what one finds when one discusses the Riemann integral of some function; we break the interval of integration in question up into small subintervals Δ_i and evaluate the function at some point x_i in each interval. Then

$$\sum_i \Delta_i f(x_i) \to \int f(x)\, dx.$$

In this spirit, as $L \to \infty$, we have for G_1 just

$$G_1(x, x', \tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x-x')} e^{-k^2\tau}\, dk. \tag{7.127}$$

It is not difficult to evaluate this integral. We note that

$$ik(x - x') - k^2\tau = -\tau\left[k - \frac{i(x - x')}{2\tau}\right]^2 - \frac{(x - x')^2}{4\tau},$$

so that

$$G_1(x, x', \tau) = \frac{1}{2\pi} e^{-(x-x')^2/4\tau} \int_{-\infty}^{\infty} e^{-\tau(k-i\delta)^2}\, dk,$$

where $\delta = (x - x')/2\tau$. To evaluate this integral, we consider the integrand to be a function of a complex variable z, and investigate the contour integral along the curve C shown in Fig. 7.10. Thus

$$I = \int_C e^{-\tau(z-i\delta)^2}\, dz = 0$$

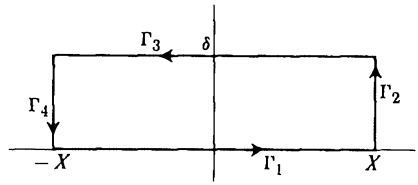

Fig. 7.10 The curve C used to evaluate the function $G_1(x, x', \tau)$. X is to be considered as being arbitrarily large.

since there are no singularities inside the contour. On Γ_1, $z = x$, and on Γ_3, $z = x + i\delta$. Thus

$$I = 0 = \int_{-X}^{X} e^{-\tau(x-i\delta)^2}\,dx + \int_{\Gamma_2} e^{-\tau(z-i\delta)^2}\,dz$$
$$+ \int_{X}^{-X} e^{-\tau x^2}\,dx + \int_{\Gamma_4} e^{-\tau(z-i\delta)^2}\,dz\,.$$

Now on Γ_2, $z = X + iy$, where X can become arbitrarily large. Hence

$$\int_{\Gamma_2} e^{-\tau(z-i\delta)^2}\,dz = i\int_0^{\delta} e^{-\tau[X+i(y-\delta)]^2}\,dy$$
$$= ie^{-\tau X^2}\int_0^{\delta} e^{\tau(y-\delta)^2}e^{-2iX\tau(y-\delta)}\,dy\,.$$

Clearly, the integral on y yields a finite number since we integrate a bounded function over a finite interval. Thus as $X \to \infty$, the integral along Γ_2 vanishes because of the factor $e^{-X^2\tau}$. The same argument eliminates the integral along Γ_4. Thus

$$\int_{-\infty}^{\infty} e^{-\tau(x-i\delta)^2}\,dx = \int_{-\infty}^{\infty} e^{-\tau x^2}\,dx = \sqrt{\frac{\pi}{\tau}}\,,$$

where this last integral can be evaluated by real integral techniques, (or see Eq. 6.56). (It should be remarked that, although we have tacitly assumed that τ is real in this calculation, it is a simple matter [see Problem 31 (l) of Chapter 6] to show that the result is valid even when τ is pure imaginary.) Thus, finally,

$$G_1(x, x', \tau) = \frac{1}{\sqrt{4\pi\tau}}\,e^{-(x-x')^2/4\tau}\,. \tag{7.128}$$

Note that $G_1(x, x', \tau)$ is a symmetric function of x and x'. As τ tends to zero,

$$\lim_{\tau\to 0} G(x, x', \tau) = \lim_{\tau\to 0} \frac{1}{\sqrt{4\pi\tau}}\,e^{-(x-x')^2/4\tau}\,,$$

which is just one of the representations of the δ-function discussed in Section 5.3. Thus

$$\lim_{\tau\to 0} G(x, x', \tau) = \delta(x - x')\,,$$

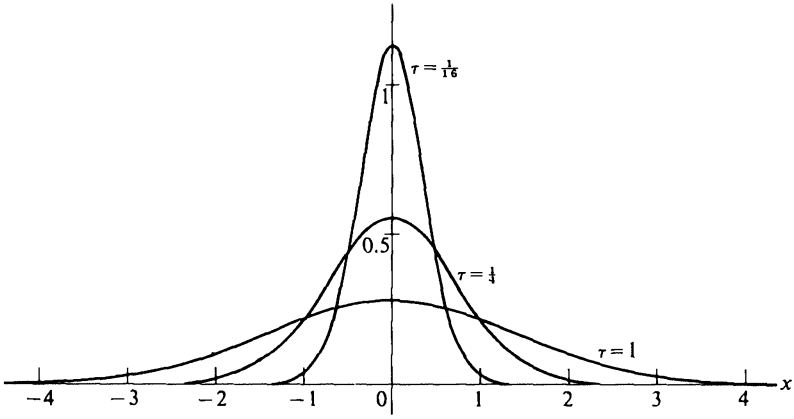

Fig. 7.11 The Green's function $G(x, x', \tau)$ as a function of x (centered at $x' = 0$ for convenience) for several values of τ.

in agreement with the general result of Eq. (7.122). As the time, τ, increases from zero, $G_1(x, x', \tau)$ looks like the familiar Gaussian, centered about $x = x'$ with half width equal to $\sqrt{2\tau}$. For small τ, the Gaussian is tall and narrow; as τ increases, the Gaussian broadens out and its height decreases, as shown in Fig. 7.11. However, the area under these Gaussians remains constant, since we see that

$$\int_{-\infty}^{\infty} G_1(x, x')\, dx = \frac{1}{\sqrt{4\pi\tau}} \int_{-\infty}^{\infty} e^{-(x-x')^2/4\tau}\, dx = 1 ,$$

independent of τ and x'. In terms of the diffusion equation, this means that the total quantity of diffusing gas (the integral of the density over all space) is constant, since

$$\psi(x, \tau) = \int_{-\infty}^{\infty} G_1(x, x', \tau)\psi(x', 0)\, dx'$$

implies that

$$\int_{-\infty}^{\infty} \psi(x, \tau)\, dx = \int_{-\infty}^{\infty} \psi(x', 0)\, dx' , \tag{7.129}$$

by the result above. In the diffusion equation, $\psi(x, \tau)$ is the density of the diffusing gas, so

$$\int_{-\infty}^{\infty} \psi(x, \tau)\, dx$$

is the total amount of diffusing gas. Equation (7.129) states that this quantity is constant in time.

In terms of this Green's function we can write the solution to the diffusion or heat equation at a later time in terms of the initial distribution:

$$\psi(x, \tau) = \frac{1}{\sqrt{4\pi\tau}} \int_{-\infty}^{\infty} e^{-(x-x')^2/4\tau} \psi(x', 0)\, dx' . \tag{7.130}$$

This equation has a simple physical interpretation: At $\tau = 0$, each element of space has an amount of diffusing material (or heat) equal to $\psi(x', 0)\, dx'$. The effect of this source at a later time at a point x is obtained by multiplying by the Green's function, which describes how the material (or heat) diffuses away from the point x'. The density (or temperature) at time τ at the point x is obtained by linear superposition of the effects at x due to all sources at the different points x'—that is, by integrating over the source.

Now suppose that we want to solve an equation of the form

$$H\psi + \frac{\partial \psi}{\partial \tau} = F(\mathbf{r}, \tau) . \tag{7.131}$$

We would like to find a Green's function such that

$$\psi(\mathbf{r}, \tau) = \psi_0(\mathbf{r}, \tau) + \int G(\mathbf{r}, \mathbf{r}', \tau, \tau') F(\mathbf{r}', \tau')\, d^3\mathbf{r}\, d\tau' , \tag{7.132}$$

where $\psi_0(\mathbf{r}, \tau)$ is a solution of the homogeneous equation. In the case of the heat equation or the diffusion equation, we might have a source of heat or a source of diffusing material which continuously varies in time described by the function $F(\mathbf{r}, \tau)$. In Schrödinger's equation, $F(\mathbf{r}, \tau)$ might represent a perturbation of some sort, such as an electromagnetic wave incident on a quantum mechanical system.

What is the relationship between the function $G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')$ discussed above and a Green's function of the type required in Eq. (7.132)? First of all, the function $G(\mathbf{r}, \mathbf{r}', \tau, \tau')$ in Eq. (7.132) must satisfy

$$\left(H + \frac{\partial}{\partial \tau}\right) G(\mathbf{r}, \mathbf{r}', \tau, \tau') = \delta^3(\mathbf{r} - \mathbf{r}')\delta(\tau - \tau') , \tag{7.133}$$

whereas G_1 satisfies

$$\left(H + \frac{\partial}{\partial \tau}\right) G_1(\mathbf{r}, \mathbf{r}', \tau, \tau') = 0 . \tag{7.133$'$}$$

Now G_1 has one particularly interesting property which was emphasized above, namely,

$$\lim_{\tau \to \tau'} G_1(\mathbf{r}, \mathbf{r}', \tau, \tau') = \delta^3(\mathbf{r} - \mathbf{r}') , \tag{7.134}$$

according to Eq. (7.122). This suggests that if we can introduce a discontinuity in G_1 at $\tau = \tau'$, we may be able to satisfy Eq. (7.133). Since G must satisfy Eq. (7.134) everywhere except at $\mathbf{r} = \mathbf{r}'$, $\tau = \tau'$, it seems reasonable to try the

modification

$$G(\mathbf{r}, \mathbf{r}', \tau, \tau') = G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')\theta(\tau - \tau'); \tag{7.135}$$

that is, for $\tau > \tau'$, $G(\mathbf{r}, \mathbf{r}', \tau, \tau')$ is just equal to $G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')$, but for $\tau < \tau'$, $G(\mathbf{r}, \mathbf{r}', \tau, \tau') = 0$. We now compute

$$\begin{aligned}
&\left(H + \frac{\partial}{\partial\tau}\right) G(\mathbf{r}, \mathbf{r}', \tau, \tau') \\
&\quad = H[G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')\theta(\tau - \tau')] + \frac{\partial}{\partial\tau}[G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')\theta(\tau - \tau')] \\
&\quad = \theta(\tau - \tau')HG_1(\mathbf{r}, \mathbf{r}', \tau, \tau') + \theta(\tau - \tau')\frac{\partial}{\partial\tau}G_1(\mathbf{r}, \mathbf{r}', \tau, \tau') \\
&\qquad + G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')\frac{\partial}{\partial\tau}\theta(\tau - \tau') \\
&\quad = \theta(\tau - \tau')\left[H + \frac{\partial}{\partial\tau}\right]G_1(\mathbf{r}, \mathbf{r}', \tau, \tau') + G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')\delta(\tau - \tau'),
\end{aligned}$$

since by Eq. (7.22),

$$\frac{\partial}{\partial\tau}\theta(\tau - \tau') = \delta(\tau - \tau').$$

Using Eq. (7.133′), we then have

$$\left(H + \frac{\partial}{\partial\tau}\right) G(\mathbf{r}, \mathbf{r}', \tau, \tau') = G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')\delta(\tau - \tau').$$

But since $\delta(\tau - \tau')$ is zero except at $\tau = \tau'$, we are only interested in $G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')$ at $\tau = \tau'$, which by Eq. (7.134) is just $\delta^3(\mathbf{r} - \mathbf{r}')$. Thus we have the desired result,

$$\left(H + \frac{\partial}{\partial\tau}\right) G(\mathbf{r}, \mathbf{r}', \tau, \tau') = \delta^3(\mathbf{r} - \mathbf{r}')\delta(\tau - \tau').$$

Using Eq. (7.135), we see that the solution to Eq. (7.131) is given by

$$\psi(\mathbf{r}, \tau) = \psi_0(\mathbf{r}, \tau) + \int d^3\mathbf{r}' \int_{-\infty}^{\tau} d\tau'\, G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')F(\mathbf{r}', \tau'). \tag{7.136}$$

The object G_1 is seen to be very nearly a Green's function, in our original sense of the word. Since $F(\mathbf{r}, \tau)$ represents the effect of a known source of heat or diffusing material, once G_1 has been determined we need only evaluate the integrals in question.

A different situation arises in quantum mechanics, where Schrödinger's time-dependent equation has the form

$$\left(H + \frac{\partial}{\partial\tau}\right)\Psi(\mathbf{r}, \tau) = -\frac{2m}{\hbar^2}U(\mathbf{r}, \tau)\Psi(\mathbf{r}, \tau);$$

in the simplest case H might be given by $H = -\nabla^2$ and $\tau = i\hbar t/2m$. Thus we obtain from Eq. (7.136) Schrödinger's time-dependent integral equation

$$\Psi(\mathbf{r}, \tau) = \Psi_0(\mathbf{r}, \tau) - \frac{2m}{\hbar^2} \int d^3\mathbf{r}' \int_{-i\infty}^{\tau} d\tau' \, G(\mathbf{r}, \mathbf{r}', \tau, \tau') U(\mathbf{r}', \tau') \Psi(\mathbf{r}', \tau') \,, \tag{7.137}$$

where the integral is along the imaginary axis, since $\tau = i\hbar t/2m$ is pure imaginary.

When $U(\mathbf{r}, \tau)$ represents a perturbation like an oscillating electromagnetic field, Eq. (7.137) leads to the theory of transition probabilities and the "Fermi Golden Rule."

We can also use Eq. (7.137) to obtain a different perspective on the work of previous sections. Suppose that $U(\mathbf{r}, \tau)$ is a static (time-independent) potential. Then we would expect to see Eq. (7.137) reduce to the Lippmann-Schwinger equation (7.68). However, rather than simply replacing $U(\mathbf{r}, \tau)$ by $V(\mathbf{r})$, let us imagine that in the very remote past $U(\mathbf{r}, \tau)$ was "turned off" and that we slowly allowed $U(\mathbf{r}, \tau)$ to be "turned on." We may express this by writing

$$U(\mathbf{r}, \tau) = V(\mathbf{r}) e^{\epsilon' t} = V(\mathbf{r}) e^{-i\epsilon\tau}; \tag{7.138}$$

by making the real quantity $\epsilon' = \hbar\epsilon/2m$ small enough, we can approximate a static potential arbitrary well. Clearly, this should have no effect on any physically realistic scattering problem. If we take $H = -\nabla^2$, as we have done before in discussing scattering, then by direct analogy with Eq. (7.127)

$$\begin{aligned} G_1(\mathbf{r}, \mathbf{r}', \tau, \tau') = (1/(2\pi)^3) \int dk_x \, dk_y \, dk_z \\ \times \exp\{i[k_x(x - x') + k_y(y - y') + k_z(z - z')]\} \\ \times \exp[-(k_x^2 + k_y^2 + k_z^2)(\tau - \tau')] \end{aligned}$$

where we imagine that we have put our system in a large rectangular box and afterwards let the sides become arbitrarily large. In spherical coordinates, this result appears as

$$G_1(\mathbf{r}, \mathbf{r}', \tau, \tau') = \frac{1}{(2\pi)^3} \int e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')} e^{-k^2(\tau-\tau')} \, d^3\mathbf{k} \,. \tag{7.139}$$

Putting Eq. (7.139) and Eq. (7.138) for $U(\mathbf{r}, \tau)$ into Eq. (7.137), we get

$$\begin{aligned} \Psi(\mathbf{r}, \tau) = \Psi_0(\mathbf{r}, \tau) \\ - (2m/\hbar^2(2\pi)^3) \int d^3\mathbf{r}' \int d^3\mathbf{k} \int_{-i\infty}^{\tau} d\tau' \, e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')} \\ \times e^{-k^2(\tau-\tau')} V(\mathbf{r}') e^{-i\epsilon\tau'} \Psi(\mathbf{r}', \tau') \end{aligned} \tag{7.140}$$

with

$$\Psi_0(\mathbf{r}, \tau) = C e^{i\mathbf{k}_i\cdot\mathbf{r}} e^{-k_i^2\tau} \,,$$

where k_i is the wave number of the incident wave. Let us look for a solution of the form

$$\Psi(\mathbf{r}, \tau) = \psi(\mathbf{r})e^{-k_i^2\tau} .$$

With this substitution, Eq. (7.140) becomes

$$\begin{aligned}\psi(\mathbf{r})e^{-k_i^2\tau} &= Ce^{i\mathbf{k}_i\cdot\mathbf{r}}e^{-k_i^2\tau} \\ &\quad - \frac{2m}{\hbar^2}\int d^3\mathbf{r}' \frac{1}{(2\pi)^3}\int d^3\mathbf{k}e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')}V(\mathbf{r}')\psi(\mathbf{r}') \\ &\quad \times e^{-k^2\tau}\int_{-i\infty}^{\tau} e^{(k^2-k_i^2-i\epsilon)\tau'}\,d\tau' \qquad (7.141)\end{aligned}$$

Now,

$$\int_{-i\infty}^{\tau} e^{(k^2-k_i^2-i\epsilon)\tau'}\,d\tau' = \frac{e^{(k^2-k_i^2-i\epsilon)\tau}}{(k^2-k_i^2-i\epsilon)} ; \qquad (7.142)$$

the lower limit of integration gives a vanishing contribution because the perturbation was switched on adiabatically via the term $\exp(-i\epsilon\tau)$. Putting Eq. (7.142) into Eq. (7.141) and canceling a common factor of $\exp(-k_i^2\tau)$, we get

$$\begin{aligned}\psi(\mathbf{r}) &= Ce^{i\mathbf{k}_i\cdot\mathbf{r}} - \frac{2m}{\hbar^2}\int d^3\mathbf{r}'\,V(\mathbf{r}')\psi(\mathbf{r}')\frac{1}{(2\pi)^3} \\ &\quad \times\int \frac{e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')}}{(k^2-k_i^2-i\epsilon)}e^{-i\epsilon\tau}\,d^3\mathbf{k} . \qquad (7.143)\end{aligned}$$

The integral

$$I(\mathbf{r}, \mathbf{r}') = -\frac{1}{(2\pi)^3}\int \frac{e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')}}{(k^2-k_i^2-i\epsilon)}\,d^3\mathbf{k}$$

is precisely the *outgoing spherical wave* Green's function $G_+(\mathbf{r}, \mathbf{r}')$ obtained for the operator $-\nabla^2 - k_i^2$ in Section 7.4 (Eq. 7.66). Thus

$$I(\mathbf{r}, \mathbf{r}') = -\frac{1}{4\pi}\frac{\exp[i(k_i + i\epsilon/2k_i)|\mathbf{r}-\mathbf{r}'|]}{|\mathbf{r}-\mathbf{r}'|} .$$

We now let $\epsilon \to 0$ and find that

$$\psi(\mathbf{r}) = Ce^{i\mathbf{k}_i\cdot\mathbf{r}} - \frac{m}{2\pi\hbar^2}\int \frac{e^{ik_i|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}V(\mathbf{r}')\psi(\mathbf{r}')\,d^3\mathbf{r}' ,$$

which agrees with the Lippmann-Schwinger equation for ψ_+ which we derived earlier, Eq. (7.68). Thus in the limit of a static potential, the time-dependent formalism goes over to the form we should expect, with the prescription we used earlier for avoiding singularities being automatically provided. We now see that this prescription can be thought of as arising from the physically reasonable assumption that at sufficiently remote times in the past there is no interaction, just a free particle.

7.8 THE WAVE EQUATION

As a final example of the Green's function technique, we investigate the Lorentz-invariant, time-dependent wave equation. This is the basic equation of electrodynamics, and its properties have been exhaustively analyzed.

The homogeneous wave equation is

$$\Box^2\psi \equiv \nabla^2\psi - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} = 0 . \tag{7.144}$$

For the sake of generality, we write this in the form

$$H\psi + \frac{\partial^2\psi}{\partial\tau^2} = 0 , \tag{7.145}$$

where, in this case, $\tau = ct$ and $H = -\nabla^2$. For the moment, let us imagine that H is a general linear operator with a complete orthonormal set of eigenfunctions ϕ_n which satisfy

$$H\phi_n = \lambda_n\phi_n . \tag{7.146}$$

Then we look for a solution of Eq. (7.145) in the form

$$\psi(\mathbf{r}, \tau) = \sum_n A_n(\tau)\phi_n(\mathbf{r}) . \tag{7.147}$$

Using Eq. (7.146) and Eq. (7.147), we find

$$H\psi + \frac{\partial^2\psi}{\partial\tau^2} = \sum_n (A_n\lambda_n + \ddot{A}_n)\phi_n = 0 . \tag{7.148}$$

Since the eigenfunctions of H form a complete orthonormal set, we conclude from Eq. (7.148) that

$$\ddot{A}_n(\tau) + \lambda_n A_n(\tau) = 0 .$$

Hence

$$A_n(\tau) = a_n e^{i\sqrt{\lambda_n}\tau} + b_n e^{-i\sqrt{\lambda_n}\tau} ,$$

and

$$\psi(\mathbf{r}, \tau) = \sum_n (a_n e^{i\sqrt{\lambda_n}\tau} + b_n e^{-i\sqrt{\lambda_n}\tau})\phi_n(\mathbf{r}) . \tag{7.149}$$

The constants a_n and b_n must be determined from the boundary conditions, which we assume to be of the familiar type; that is, $\psi(\mathbf{r}, 0)$ and $\dot{\psi}(\mathbf{r}, 0)$ are specified. Then

$$\psi(\mathbf{r}, 0) = \sum_n (a_n + b_n)\phi_n(\mathbf{r}) ,$$

$$\dot{\psi}(\mathbf{r}, 0) = i\sum_n \sqrt{\lambda_n}(a_n - b_n)\phi_n(\mathbf{r}) .$$

We find that

$$a_n + b_n = \int \phi_n^*(\mathbf{r}')\psi(\mathbf{r}', 0)\, d^3\mathbf{r}' ,$$

$$a_n - b_n = -\frac{i}{\sqrt{\lambda_n}} \int \phi_n^*(\mathbf{r}')\dot{\psi}(\mathbf{r}', 0)\, d^3\mathbf{r}' ,$$

so

$$a_n = \tfrac{1}{2}\left[\int \phi_n^*(\mathbf{r}')\psi(\mathbf{r}', 0)\, d^3\mathbf{r}' + \frac{1}{i\sqrt{\lambda_n}} \int \phi_n^*(\mathbf{r}')\dot{\psi}(\mathbf{r}', 0)\, d^3\mathbf{r}'\right],$$

$$b_n = \tfrac{1}{2}\left[\int \phi_n^*(\mathbf{r}')\psi(\mathbf{r}', 0)\, d^3\mathbf{r}' - \frac{1}{i\sqrt{\lambda_n}} \int \phi_n^*(\mathbf{r}')\dot{\psi}(\mathbf{r}', 0)\, d^3\mathbf{r}'\right].$$

Inserting these values for a_n and b_n into Eq. (7.149), we get

$$\psi(\mathbf{r}, \tau) = \int \sum_n \cos(\sqrt{\lambda_n}\tau)\phi_n(\mathbf{r})\phi_n^*(\mathbf{r}')\psi(\mathbf{r}', 0)\, d^3\mathbf{r}'$$
$$+ \int \sum_n \sin(\sqrt{\lambda_n}\tau)\frac{1}{\sqrt{\lambda_n}}\phi_n(\mathbf{r})\phi_n^*(\mathbf{r}')\dot{\psi}(\mathbf{r}', 0)\, d^3\mathbf{r}' .$$

In keeping with the notation of the previous section, let us write this as

$$\psi(\mathbf{r}, \tau) = \int G_2(\mathbf{r}, \mathbf{r}', \tau)\psi(\mathbf{r}', 0)\, d^3\mathbf{r}' + \int \tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau)\dot{\psi}(\mathbf{r}', 0)\, d^3\mathbf{r}' , \qquad (7.150)$$

where

$$G_2(\mathbf{r}, \mathbf{r}', \tau) = \sum_n \cos(\sqrt{\lambda_n}\tau)\phi_n(\mathbf{r})\phi_n^*(\mathbf{r}') , \qquad (7.151a)$$

$$\tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau) = \sum_n \frac{\sin\sqrt{\lambda_n}\tau}{\sqrt{\lambda_n}}\phi_n(\mathbf{r})\phi_n^*(\mathbf{r}') . \qquad (7.151b)$$

Since the wave equation is second order in time, two boundary conditions have been required to determine the constants of integration. Hence we have obtained two Green's functions, one which acts on $\psi(\mathbf{r}, 0)$, the other which acts on $\dot{\psi}(\mathbf{r}, 0)$. For the first-order (in time) heat or diffusion equation, only one boundary condition is needed, and hence we found only one Green's function which acted on $\psi(\mathbf{r}, 0)$. Just as the Green's function obtained in the last section satisfied the heat equation, so both of our Green's functions, G_2 and $\tilde{G}_2$, satisfy the wave equation;

$$\Box^2 G_2(\mathbf{r}, \mathbf{r}', \tau) = 0 , \qquad \Box^2 \tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau) = 0 .$$

Thus $\psi(\mathbf{r}, \tau)$ satisfies the wave equation. Note also that

$$G_2(\mathbf{r}, \mathbf{r}', \tau) = \frac{d}{d\tau}\tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau) ,$$

so that if we can calculate $\tilde{G}_2$, then G_2 can be obtained directly from it. The boundary conditions are clearly satisfied by the solution given in Eq. (7.150) since

$$G_2(\mathbf{r}, \mathbf{r}', 0) = \sum_n \phi_n(\mathbf{r})\phi_n^*(\mathbf{r}') = \delta^3(\mathbf{r} - \mathbf{r}') . \tag{7.152}$$

With these general results in mind, let us return to the specific case of the wave equation, and start by considering a region defined by a rectangular parallelepiped with sides L_1, L_2, and L_3 in the x, y- and z-directions, respectively. If we have periodic boundary conditions, the eigenfunctions of $H = -\nabla^2$ may be written as

$$\phi_{n_1 n_2 n_3}(\mathbf{r}) = \frac{1}{\sqrt{L_1 L_2 L_3}} e^{ik_x x} e^{ik_y y} e^{ik_z z} ,$$

where

$$k_x = \frac{2\pi n_1}{L_1} , \qquad k_y = \frac{2\pi n_2}{L_2} , \qquad k_z = \frac{2\pi n_3}{L_3} ,$$

and the eigenvalues are given simply by

$$\lambda_{n_1 n_2 n_3} = k_x^2 + k_y^2 + k_z^2 .$$

Let us now write $\tilde{G}_2$ in terms of these quantities. According to Eq. (7.151b),

$$\tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau) = \frac{1}{L_1 L_2 L_3} \sum_{n_1, n_2, n_3}^{\infty} \frac{\sin \sqrt{k_x^2 + k_y^2 + k_z^2}\,\tau}{\sqrt{k_x^2 + k_y^2 + k_z^2}} e^{ik_x(x-x')} e^{ik_y(y-y')} e^{ik_z(z-z')} .$$

Now we can move to the limit in which the box becomes arbitrarily large, as we did before. Then

$$\frac{1}{L_1} \sum_{n_1=-\infty}^{\infty} \to \frac{1}{2\pi} \int_{-\infty}^{\infty} dk_x$$

and similarly for the sums on n_2 and n_3. Hence, in the limit of very large dimensions,

$$\tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau) = \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} dk_x \int_{-\infty}^{\infty} dk_y \int_{-\infty}^{\infty} dk_z \frac{\sin k\tau}{k} e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')} ,$$

where $\mathbf{k} = k_x\mathbf{i} + k_y\mathbf{j} + k_z\mathbf{k}$ and $k^2 = k_x^2 + k_y^2 + k_z^2$. It will clearly be advantageous to evaluate this integral by using spherical coordinates. Setting $\mathbf{r} - \mathbf{r}' \equiv \boldsymbol{\rho}$, we get

$$\tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau) = \frac{1}{(2\pi)^3} \int_0^{\infty} dk \cdot k^2 \sin k\tau \cdot \frac{1}{k} \int_0^{\pi} \sin\theta_k \, d\theta_k \int_0^{2\pi} d\phi_k e^{i\mathbf{k}\cdot\boldsymbol{\rho}} .$$

Since the integral is independent of the direction of $\boldsymbol{\rho}$, let us for convenience choose $\boldsymbol{\rho}$ to be in the z-direction. Then $\mathbf{k}\cdot\boldsymbol{\rho} = k\rho\cos\theta_k$, and making the

change of variable $x = \cos\theta_k$, we have

$$\begin{aligned}\tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau) &= \frac{1}{(2\pi)^3}\int_0^\infty dk\cdot k \sin k\tau \int_{-1}^{1} dx \int_0^{2\pi} d\phi_k e^{ik\rho x} \\ &= \frac{1}{2\pi^2\rho}\int_0^\infty \sin k\tau \sin k\rho \, dk \\ &= \frac{1}{4\pi^2\rho}\int_{-\infty}^\infty \sin k\tau \sin k\rho \, dk \\ &= \frac{1}{8\pi^2\rho}\int_{-\infty}^\infty [\cos k(\rho - \tau) - \cos k(\rho + \tau)]\, dk \,.\end{aligned}$$

Since

$$\delta(x) = \frac{1}{2\pi}\int_{-\infty}^\infty \cos xy \, dy$$

(this follows immediately from Eq. 5.63), we get finally

$$\tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau) = \frac{1}{4\pi\rho}[\delta(\rho - \tau) - \delta(\rho + \tau)] \,.$$

In terms of the original variables, this is

$$\tilde{G}_2(\mathbf{r}, \mathbf{r}', t) = \frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|}[\delta(|\mathbf{r} - \mathbf{r}'| - ct) - \delta(|\mathbf{r} - \mathbf{r}'| + ct)] \,. \tag{7.153}$$

Clearly, our choice of $t = 0$ as the initial time was arbitrary. The reader can check that if we choose t' as the initial time, then

$$\begin{aligned}\psi(\mathbf{r}, t) &= \int G_2(\mathbf{r}, \mathbf{r}', t, t')\psi(\mathbf{r}', t')\, d^3\mathbf{r}' \\ &\quad + \int \tilde{G}_2(\mathbf{r}, \mathbf{r}', t, t')\dot{\psi}(\mathbf{r}', t')\, d^3\mathbf{r}' \,,\end{aligned} \tag{7.154}$$

where

$$\tilde{G}_2(\mathbf{r}, \mathbf{r}', t, t') = \sum_n \frac{\sin[\sqrt{\lambda_n}c(t - t')]}{\sqrt{\lambda_n}}\phi_n(\mathbf{r})\phi_n^*(\mathbf{r}') \,, \tag{7.155a}$$

and

$$\begin{aligned}G_2(\mathbf{r}, \mathbf{r}', t, t') &= \sum_n \cos[\sqrt{\lambda_n}c(t - t')]\phi_n(\mathbf{r})\phi_n^*(\mathbf{r}') \\ &= \frac{1}{c}\frac{d}{dt}\tilde{G}_2(\mathbf{r}, \mathbf{r}', t, t') \,,\end{aligned} \tag{7.155b}$$

For the case of the wave equation, $H = -\nabla^2$, $\tilde{G}_2$ is given by

$$\begin{aligned}\tilde{G}_2(\mathbf{r}, \mathbf{r}', t, t') = \frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|}\{&\delta[|\mathbf{r} - \mathbf{r}'| - c(t - t')] \\ &- \delta[|\mathbf{r} - \mathbf{r}'| + c(t - t')]\} \,.\end{aligned} \tag{7.156}$$

This gives the solution to the homogeneous wave equation, once the boundary conditions are specified.

However, just as in the case of the first-order equation studied in the last section, the inhomogeneous equation is very important. In classical electrodynamics, for example, Eq. (7.144) corresponds to a situation where no sources or currents are present. However, in general, one is interested in cases where there are currents and charges present. Thus, typically, we would like to be able to solve

$$\Box^2\psi \equiv \nabla^2\psi - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} = \frac{4\pi}{c} j(\mathbf{r}, t) , \tag{7.157}$$

or, in general,

$$H\psi + \frac{\partial^2\psi}{\partial\tau^2} = J(\mathbf{r}, \tau) . \tag{7.158}$$

Now, to solve Eq. (7.158), we need a Green's function $G(\mathbf{r}, \mathbf{r}', \tau, \tau')$ which satisfies

$$\left(H + \frac{\partial^2}{\partial\tau^2}\right) G(\mathbf{r}, \mathbf{r}', \tau, \tau') = \delta^3(\mathbf{r} - \mathbf{r}')\delta(\tau - \tau') .$$

Neither G_2 nor $\tilde{G}_2$ has this property; in fact, letting $\mathscr{G}$ stand for either G_2 or $\tilde{G}_2$,

$$\left(H + \frac{\partial^2}{\partial\tau^2}\right) \mathscr{G} = 0 . \tag{7.159}$$

However, let us be guided by our experience in the previous section and look for G in the form

$$G_R(\mathbf{r}, \mathbf{r}', \tau, \tau') = \mathscr{G}(\mathbf{r}, \mathbf{r}', \tau, \tau')\theta(\tau - \tau') .$$

We have

$$\begin{aligned}
&\left(H + \frac{\partial^2}{\partial\tau^2}\right) G_R(\mathbf{r}, \mathbf{r}', \tau, \tau') \\
&\quad = \theta(\tau - \tau')H\mathscr{G} + \frac{\partial^2}{\partial\tau^2}[\mathscr{G}\theta(\tau - \tau')] \\
&\quad = \theta(\tau - \tau')\left(H + \frac{\partial^2}{\partial\tau^2}\right)\mathscr{G} + 2\dot{\theta}(\tau - \tau')\dot{\mathscr{G}} + \mathscr{G}\ddot{\theta}(\tau - \tau') ,
\end{aligned}$$

where a dot indicates differentiation with respect to τ. By Eq. (7.159), this is just

$$\left(H + \frac{\partial^2}{\partial\tau^2}\right) G_R = 2\dot{\theta}(\tau - \tau')\dot{\mathscr{G}} + \mathscr{G}\ddot{\theta}(\tau - \tau') .$$

But $\dot{\theta}(\tau - \tau') = \delta(\tau - \tau')$, so

$$\begin{aligned}
&\left(H + \frac{\partial^2}{\partial\tau^2}\right) G_R(\mathbf{r}, \mathbf{r}', \tau, \tau') \\
&\quad = 2\delta(\tau - \tau')\dot{\mathscr{G}}(\mathbf{r}, \mathbf{r}', \tau, \tau') + \dot{\delta}(\tau - \tau')\mathscr{G}(\mathbf{r}, \mathbf{r}', \tau, \tau') ,
\end{aligned}$$

From the results of Section 7.2,

$$\delta(\tau - \tau')\,\dot{\mathscr{G}}(\mathbf{r}, \mathbf{r}', \tau, \tau') = [\,\dot{\mathscr{G}}(\mathbf{r}, \mathbf{r}', \tau, \tau')]_{\tau'=\tau}\delta(\tau - \tau')\ ,$$
$$\dot{\delta}(\tau - \tau')\,\mathscr{G}(\mathbf{r}, \mathbf{r}', \tau, \tau') = -[\,\dot{\mathscr{G}}(\mathbf{r}, \mathbf{r}', \tau, \tau')]_{\tau'=\tau}\delta(\tau - \tau')\ .$$

Thus

$$\left(H + \frac{\partial^2}{\partial\tau^2}\right) G_R(\mathbf{r}, \mathbf{r}', \tau, \tau') = \delta(\tau - \tau')[\,\dot{\mathscr{G}}(\mathbf{r}, \mathbf{r}', \tau, \tau')\,]_{\tau'=\tau}\ . \tag{7.160}$$

If $\mathscr{G} = G_2$, the right-hand side of Eq. (7.160) vanishes by Eq. (7.155b), but if $\mathscr{G} = \tilde{G}_2$, then according to Eq. (7.155a) we get

$$\left(H + \frac{\partial^2}{\partial\tau^2}\right) G_R(\mathbf{r}, \mathbf{r}', \tau, \tau') = \delta(\tau - \tau')\delta^3(\mathbf{r} - \mathbf{r}')\ ,$$

which is the desired form. Hence the Green's function for the homogeneous equation is

$$G_R(\mathbf{r}, \mathbf{r}', \tau, \tau') = \theta(\tau - \tau')\tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau, \tau')\ , \tag{7.161}$$

and the solution to Eq. (7.158) can be written as

$$\psi_R(\mathbf{r}, \tau) = \psi_0(\mathbf{r}, \tau) + \int d^3\mathbf{r}' \int_{-\infty}^{\tau} d\tau'\, \tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau, \tau')J(\mathbf{r}', \tau')\ , \tag{7.162}$$

where $\psi_0(\mathbf{r}, \tau)$ is a solution of the homogeneous equation.

Now for the special case of the wave equation, $\tilde{G}_2$ is given by Eq. (7.156); therefore, since G_R vanishes for $\tau < \tau'$, we may write

$$G_R(\mathbf{r}, \mathbf{r}', \tau, \tau') = \frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|}\,\delta(|\mathbf{r} - \mathbf{r}'| - (\tau - \tau'))\theta(\tau - \tau')\ ,$$

for when $\tau > \tau'$ the argument of the second δ-function in Eq. (7.156) is always nonzero. Since $\delta(|\mathbf{r} - \mathbf{r}'| - (\tau - \tau'))$ has zero argument only when $\tau > \tau'$, we may drop the factor $\theta(\tau - \tau')$ from G_R so that finally

$$G_R(\mathbf{r}, \mathbf{r}', \tau, \tau') = \frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|}\,\delta(|\mathbf{r} - \mathbf{r}'| - (\tau - \tau'))\ . \tag{7.163}$$

Thus the solution to the wave equation is

$$\psi_R(\mathbf{r}, \tau) = \psi_0(\mathbf{r}, \tau) + \frac{1}{4\pi}\int d^3\mathbf{r}' \int_{-\infty}^{\infty} d\tau'\, \frac{1}{|\mathbf{r} - \mathbf{r}'|}\,\delta(|\mathbf{r} - \mathbf{r}'| - (\tau - \tau'))J(\mathbf{r}', \tau')\ .$$

The τ integration can be done immediately, giving

$$\psi_R(\mathbf{r}, \tau) = \psi_0(\mathbf{r}, \tau) + \frac{1}{4\pi}\int \frac{[J(\mathbf{r}', \tau')]_{\text{ret}}}{|\mathbf{r} - \mathbf{r}'|}\, d^3\mathbf{r}'$$

where the bracket $[\quad]_{\text{ret}}$ means that the time τ' is to be evaluated at the *retarded*

time

$$\tau' = \tau - |\mathbf{r} - \mathbf{r}'| .$$

Writing this in terms of the time t, we have

$$\phi_R(\mathbf{r}, t) = \phi_0(\mathbf{r}, t) + \frac{1}{4\pi} \int \frac{[J(\mathbf{r}', t')]_{\text{ret}}}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}' , \tag{7.164}$$

where in the bracket

$$t' = t - \frac{|\mathbf{r} - \mathbf{r}'|}{c} .$$

This is one of the most beautiful results in electromagnetic theory. It tells us that an observer at the field point $\mathbf{r}$ is affected at a given time t only by what a source at $\mathbf{r}'$ was doing at an earlier time t'; furthermore, this earlier time t' precedes the present time t by just the amount of time it takes a wave to travel from the source point $\mathbf{r}'$ to the field point $\mathbf{r}$, namely, $|\mathbf{r} - \mathbf{r}'|/c$. In general, an effect at one point in space-time is due to all disturbances throughout space which preceded the effect by exactly the amount of time necessary for the disturbances to propagate from their points of origin to the point where their effects are felt. ϕ_R is referred to as the *retarded* solution to the inhomogeneous wave equation and G_R is called the *retarded* Green's function: hence the subscript R.

This is by no means the only Green's function that we could have defined. For example, we could write

$$G_A(\mathbf{r}, \mathbf{r}', \tau, \tau') = -\theta(\tau' - \tau)\tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau, \tau') .$$

We leave it to the reader to show that this is also a Green's function for the inhomogeneous wave equation (called the *advanced* Green's function), and to show that the solution obtained from G_A corresponds to the effects of disturbances propagating back from the future to the present.

PROBLEMS

1. Show that the Green's functions of Eqs. (7.9′) and (7.11′) satisfy

$$\frac{d^2G(x, x')}{dx^2} = \delta(x - x') .$$

2. Show that

$$\frac{d\delta(x)}{dx} = -\delta(x)\frac{d}{dx}$$

when both elements of the equation are considered to operate on a function in an integrand; i.e., we really mean

$$\int_{-a}^{b} \frac{d\delta(x)}{dx} f(x)\, dx = -\int_{-a}^{b} \delta(x) \frac{df}{dx}\, dx$$

for any $f(x)$ with a continuous derivative.

3. Show that if H is a real, Hermitian operator (that is, $H = H^*$, $H = H^\dagger$) then the Green's function for the inhomogeneous equation

$$H\phi(x) = f(x)$$

is symmetric, that is, $G(x, x') = G(x', x)$.
[*Hint*: Show that the eigenfunctions of a real, Hermitian operator can always be chosen to be real.]

4. Show that the Green's function of Eq. (7.35) satisfies Eq. (7.36).

5. In Section 6.9 we discussed those solutions of Bessel's equation,

$$x^2 \frac{d^2 J_n}{dx^2} + x \frac{dJ_n}{dx} + (x^2 - n^2) J_n = 0 ,$$

which were finite at the origin. In particular, we chose a normalization such that for small x

$$J_n(x) \cong \frac{1}{n!}\left(\frac{x}{2}\right)^n .$$

There is, of course, another set of solutions, usually denoted by $N_n(x)$, which are irregular at the origin. They are conventionally normalized so that

$$N_0(x) \cong \frac{2}{\pi} \ln x , \qquad N_n(x) \sim -\frac{1}{\pi}\left(\frac{2}{x}\right)^n (n-1)!$$

(Note that for small values of x these $N_n(x)$ do indeed satisfy Bessel's equation.)

Consider the inhomogeneous Bessel's equation (for $n \geq 1$)

$$x^2 \frac{d^2 u}{dx^2} + x \frac{du}{dx} + (x^2 - n^2) u = f(x)$$

on the interval [0, 1] with boundary condition $u(0) = 0 = u(1)$. Show that $u(x)$ is given by

$$u(x) = \int_0^1 G(x, x') f(x')\, dx ,$$

where

$$G(x, x') = -\frac{\pi}{2x' J_n(1)} J_n(x_<)[J_n(x_>) N_n(1) - N_n(x_>) J_n(1)] .$$

As usual, $x_>$ is the greater of x and x', and $x_<$ is the lesser of x and x'.

6. Prove that the Wronskian of a second-order linear differential operator never vanishes. Do this by first proving that $W(x)$ cannot be identically equal to zero (because if it were, u_1 and u_2 would be linearly dependent, contrary to assumption). Then, differentiating $W(x)$ and using the fact that u_1 and u_2 are solutions to

$$f_0(x) \frac{d^2 u}{dx^2} + f_1(x) \frac{du}{dx} + f_2(x) u = 0 ,$$

prove that

$$\frac{dW}{dx} = -\frac{f_1}{f_0} W .$$

Integrate this and show that if $f_0(x)$ and $f_1(x)$ are continuous and if $f_0(x)$ never vanishes, then $W(x)$ *never* vanishes.

7. Prove that for *any* linear differential operator of the form

$$f_0(x)\frac{d^n}{dx^n} + f_1(x)\frac{d^{n-1}}{dx^{n-1}} + \cdots + f_n(x)\,,$$

we have

$$\frac{dW}{dx} = -\frac{f_1}{f_0}W\,,$$

where W is defined by

$$W(x) = \begin{vmatrix} u_1(x) & u_2(x) & \cdots & u_n(x) \\ u_1^{(1)}(x) & u_2^{(1)}(x) & \cdots & u_n^{(1)}(x) \\ u_1^{(n-1)}(x) & u_2^{(n-1)}(x) & \cdots & u_n^{(n-1)}(x) \end{vmatrix},$$

$u_1(x), u_2(x), \cdots, u_n(x)$ are the linearly independent solutions of the equations and $u^{(m)}$ denotes the mth derivative of u. Thus $W(x)$ is given by

$$W(x) = C\exp\left[-\int\frac{f_1(x)}{f_0(x)}\,dx\right].$$

8. Show that the Green's function for the differential operator

$$f_0(x)\frac{d^n}{dx^n} + f_1(x)\frac{d^{n-1}}{dx^{n-1}} + \cdots + f_n(x)$$

is given by

$$G(x, x') = \sum_{m=1}^{n} a_m(x')u_m(x) + \frac{N(x, x')}{f_0(x')W(x')}\theta(x - x')\,,$$

where $W(x')$ is defined as in the previous problem and

$$N(x, x') = \begin{vmatrix} u_1(x') & u_2(x') & \cdots & u_n(x') \\ u_1^{(1)}(x') & u_2^{(1)}(x') & & u_n^{(1)}(x') \\ \vdots & \vdots & & \vdots \\ u_1^{(n-2)}(x') & u_2^{(n-2)}(x') & & u_n^{(n-2)}(x') \\ u_1(x) & u_2(x) & \cdots & u_n(x) \end{vmatrix}.$$

The $a_m(x')$ are determined by the boundary condition which the Green's function must satisfy. As in Problem 7, $u_1, u_2, \cdots, u_n$ are the n linearly independent solutions of the homogeneous equation.

9. Using the results of the previous problem, find the Green's function for the general third-order linear differential operator,

$$\frac{d^3}{dx^3} + a_2\frac{d^2}{dx^2} + a_2\frac{d}{dx} + a_3$$

in terms of the roots, r_1, r_2 and r_3 of the characteristic polynomial

$$z^3 + a_1z^2 + a_2z + a_3 = 0\,.$$

Assume that the Green's function and its first two derivatives vanish at $x = 0$.

10. Schrödinger's equation in one dimension is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi(x) = E\psi(x)\,.$$

Using the methods discussed in Section 7.4, derive a Lippmann-Schwinger integral equation for $\psi(x)$,

a) for the case $E < 0$, b) for the case $E \geq 0$.

In case (b) you will find separate equations for $\psi_+(x)$ and $\psi_-(x)$ corresponding to different types of boundary conditions at infinity. Discuss the meaning of $\psi_+(x)$ and $\psi_-(x)$.

Solve Problem 10 by using the method of Section 7.3. What is the appropriate statement of boundary conditions to specify the Green's function completely in this case?

a) By making the change of variable $\phi_l(x) = f_l(x)/\sqrt{x}$ in Eq. (7.81), show that the following equation is obtained for $f_l(x)$:

$$x^2 f_l''(x) + x f_l'(x) + [x^2 - (l + 1/2)^2] f_l(x) = 0.$$

b) Using the results of Section 6.9, show that

$$j_l(x) \propto \frac{1}{\sqrt{x}} J_{l+1/2}(x) ,$$

where J denotes a Bessel function [see the discussion following Eq. (7.88b′)].

. a) Show that the Green's function

$$G(t, t') = \frac{1}{2\omega_0 i} e^{i\omega_0 |t - t'|}$$

satisfies

$$\left(\frac{d^2}{dt^2} + \omega_0^2\right) G(t, t') = \delta(t - t') .$$

This Green's function was used in Problem 7.10 to convert Schrödinger's differential equation into the Lippmann-Schwinger integral equation.

b) Show that a general solution to the equation

$$X(t) + \omega_0^2 X(t) = F(t)$$

can be written in terms of the function $G(t, t')$ above as

$$X(t) = A \cos \omega_0 t + B \sin \omega_0 t + \int_{-\infty}^{\infty} G(t, t') F(t')\, dt' ,$$

where A and B are to be determined in terms of the boundary conditions. For classical mechanical one-point boundary conditions, they will clearly depend on F. Show that if we require $x(0) = x_0$ and $\dot{x}(0) = 0$, then the equation reduces to the familiar result

$$X(t) = x_0 \cos \omega_0 t + \frac{1}{\omega_0} \int_0^t \sin \omega_0 (t - t') F(t')\, dt' .$$

Clearly, for one-point boundary conditions, it is easier to start with $(1/\omega_0) \times \theta(t - t') \sin \omega_0 (t - t')$ as Green's function, but it is not necessary to do so.

4. Show from Eq. (7.98) that

$$e^{i\mathbf{k}\cdot\mathbf{r}} = \sum_{l=0}^{\infty} i^l (2l + 1) j_l(kr) P_l(\cos \theta) ,$$

where θ is the angle between $\mathbf{k}$ and $\mathbf{r}$.

[*Hint*: Consider Eq. (7.98) in the limit $|\mathbf{r}'| \to \infty$.]

15. Making use of Eqs. (7.98), (7.86), (7.84a) and (5.83), show that if we choose

$$\xi(\mathbf{r}) = \frac{A}{(2\pi)^{3/2}} e^{i\mathbf{k}\cdot\mathbf{r}} - \frac{4imk}{\hbar^2} \sum_{l=0}^{\infty} \sum_{m=-\infty}^{\infty} j_l(kr) Y_{lm}(\theta, \phi) \int j_l(kr') Y_{lm}^*(\theta', \phi') V(\mathbf{r}') \psi(\mathbf{r}')\, d\mathbf{r}' ,$$

then the equation

$$\psi(\mathbf{r}) = \xi(\mathbf{r}) - \frac{m}{2\pi\hbar^2} \int \frac{e^{-ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} V(\mathbf{r}') \psi(\mathbf{r}')\, d\mathbf{r}'$$

reduces to

$$\psi(\mathbf{r}) = \frac{A}{(2\pi)^{3/2}} e^{i\mathbf{k}\cdot\mathbf{r}} - \frac{m}{2\pi\hbar^2} \int \frac{e^{+ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} V(\mathbf{r}') \psi(\mathbf{r}')\, d\mathbf{r}' .$$

Note that $\xi(\mathbf{r})$ is a linear superposition of solutions to $(\nabla^2 + k^2)\phi = 0$.

16. In obtaining the Lippmann-Schwinger bound-state equation by the Fourier transform method, it almost appears that we applied *no* boundary conditions at all (no "$i\epsilon$-prescription" was necessary) and yet we got a uniquely determined Green's function. Show that this is not actually the case, i.e., that we really did insert a boundary condition. How was this done? Relate your discussion to the work in Section 7.5.

17. Consider a thin ring of radius R. Initially, at $t = 0$, there is a temperature distribution $T(\phi, 0)$ around the ring, where ϕ is the appropriate angular coordinate. Using the Green's function method, find the temperature at later times. Explain any decisions you make concerning boundary conditions. Suppose that the ring is initially heated in a very small region around $\phi = \phi_0$, so that the form $T(\phi, 0) = T_0\delta(\phi - \phi_0)$ is a reasonable boundary condition. Find the temperature at later times, $t > 0$. What is the value of the uniform temperature which the ring finally attains? How is the rate of approach to equilibrium affected by increasing the radius of the ring?

18. a) Consider a one-dimensional rod of length L. Let one end, at $x = 0$, be held at zero temperature and the other end, at $x = L$, be held at temperature T_0. If the initial temperature distribution is $T(x, 0)$, show that the temperature at subsequent times is given by

$$T(x, t) = T_0 x/L + \int_0^L G_1(x, x', t)\, \tilde{T}(x', 0)\, dx' ,$$

where

$$\tilde{T}(x, 0) = T(x, 0) - T_0 x/L$$

and

$$G_1(x, x', t) = \frac{2}{L} \sum_{m=1}^{\infty} \sin\frac{m\pi x}{L} \sin\frac{m\pi x'}{L} e^{(-k/c)(m\pi/L)^2 t} .$$

[*Hint*: Note that $\tilde{T}(x, t) = T(x, t) - T_0 x/L$ also satisfies the heat equation. What are the boundary conditions satisfied by $\tilde{T}(x, t)$?

b) Using the result of part (*a*), find the Green's function which will enable one to solve the equation

$$\nabla^2 T = \frac{c}{k}\frac{\partial T}{\partial t}$$

on the infinite half-line $[0, \infty]$ when $T(x, 0)$ is given some prescribed functional value.

19. a) Using the results of the previous problem (or any other method), find in terms of a Green's function the solution to the heat equation for a one-dimensional rod whose ends at $x = -L/2$ and $x = L/2$ are kept at temperatures T_1 and T_2 respectively.

b) Find an expression for the Green's function of part (a) in the limit $L \to \infty$, that is, for the infinite line. Compare this expression with the result found in Section 7.7 for periodic boundary conditions.

20. Using the eigenfunction expansion method, show that the potential energy in a two-dimensional, charge-free region bounded by circles of radii R_1 and R_2 is given by

$$V(r, \theta) = \int_0^{2\pi} G_1(\theta, \theta', r) V(R_1, \theta')\, d\theta' + \int_0^{2\pi} G_2(\theta, \theta', r) V(R_2, \theta')\, d\theta' ,$$

where $V(R_1, \theta)$ is the known value of the potential on the first surface as a function of θ, and $V(R_2, \theta)$ is its value on the second surface. Find the eigenfunction expansion of G_1 and G_2, using the fact that $\partial^2/\partial\theta^2$ is an Hermitian operator if one imposes periodic boundary conditions on $[0, 2\pi]$. Why do we require such conditions in this problem? Does the limit $V(R_1, \theta) = \text{const}$, $V(R_2, \theta) = \text{const}$ give the result you would expect? Show that $G_1(\theta, \theta', R_1) = \delta(\theta - \theta')$ and $G_2(\theta, \theta', R_2) = \delta(\theta - \theta')$.

21. a) Determine the Green's function, $G_1(x, y, x', y', t)$, for the homogeneous heat equation for the entire xy-plane (i.e., the two-dimensional Green's function for an infinite space).

b) Assume that at $t = 0$ the temperature is a constant T_0 inside a square of side $2a$, centered at the origin, and zero outside this square. Determine the temperature distribution throughout the plane for all later times t in terms of the error function:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2}\, du .$$

22. Show that the Green's function for the two-dimensional Laplacian over the entire two-dimensional space is

$$G(x, y; x', y') = +\frac{1}{2\pi} \log\left[(x - x')^2 + (y - y')^2\right]^{1/2} .$$

[*Hint*: One can always add a constant to the Green's function and it will still satisfy $\nabla^2 G = \delta(x - x')\delta(y - y')$.]

23. Show that if the wave function ψ of a particle satisfies Schrödinger's equation,

$$H\psi = ih\frac{\partial \psi}{\partial t} .$$

where H is a self-adjoint operator, and if

$$\int \psi^*(\mathbf{r}, 0)\psi(\mathbf{r}, 0)\, d^3r = 1 \tag{1}$$

at $t = 0$, then at any subsequent time t, we will also have

$$\int \psi^*(\mathbf{r}, t)\psi(\mathbf{r}, t) d^3r = 1\,. \tag{2}$$

Since in quantum mechanics $|\psi|^2$ is the probability density, $|\psi|^2 d^3r$ gives the probability of finding the particle in the volume element d^3r, and Eqs. (1) and (2) are a statement of the conservation of total probability. That is, if the total probability of finding the particle *someplace* is initially equal to unity [Eq. (1)], it remains equal to unity at all later times. Clearly, the interpretation of $|\psi|^2$ as a probability density would not make sense if this were not the case. Note that such an interpretation would not hold for solutions to the heat (or diffusion) equation.

24. Use Green's theorem to obtain Eq. (7.117) from Eq. (7.116).

25. a) Show that the function

$$G'(\mathbf{r}, \mathbf{r}', \tau, \tau') = -\, G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')\theta(\tau' - \tau)$$

is a Green's function for Eq. (7.131), where G_1 satisfies Eqs. (7.134) and (7.135).

b) This means that we can write a time-dependent Schrödinger integral equation as

$$\psi(\mathbf{r}, t) = \psi_0(\mathbf{r}, t) - \frac{2m}{\hbar^2}\int d^3r' \int_\tau^{i\infty} G'(\mathbf{r}, \mathbf{r}', \tau, \tau')\, U(\mathbf{r}', \tau')\psi(\mathbf{r}', \tau')\, d\tau'$$

in the notation of Eq. (7.137). Show that U is a static potential which is "turned off" at remote *future* times, that is,

$$U(\mathbf{r}, \tau) = V(\mathbf{r})e^{-\epsilon' t} = V(\mathbf{r})e^{i\epsilon\tau}\,,$$

then we obtain in the limit $\epsilon \to 0$ the Lippmann-Schwinger equation with an *incoming* spherical wave boundary condition solution.

26. a) Note that Eq. (7.125b) does not make sense if $\tau' > \tau$ (assuming, as will usually be the case, that $\lambda_m \to +\infty$ as $m \to \infty$) and τ and τ' are real. This is also reflected in the discussion following Eq. (7.128), where we are tacitly assuming that $\tau \to 0$ from "the positive side of zero" (see Section 5.3). Thus the Green's function

$$G(\mathbf{r}, \mathbf{r}', \tau, \tau') = -\, \theta(\tau' - \tau)G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')$$

obtained in the previous problem, although it formally seems to satisfy

$$\left(H + \frac{\partial}{\partial\tau}\right)G = \delta^3(\mathbf{r} - \mathbf{r}')\delta(\tau - \tau')\,,$$

is *not* an acceptable Green's function. Show that if one derives $G(\mathbf{r}, \mathbf{r}', \tau, \tau')$ for the case of the diffusion (or heat) equation by using the Fourier transform method, one obtains *only*

$$G(\mathbf{r}, \mathbf{r}', \tau, \tau') = \theta(\tau - \tau')G_1(\mathbf{r}, \mathbf{r}', \tau, \tau')\,,$$

where G_1 is given by Eq. (7.139).

b) For the case of Schrödinger's equation for a free particle, the expontial in Eq. (7.125b) is oscillatory because τ is imaginary. In this case, show that the Fourier

transform method yields two Green's functions:

$$G_+(\mathbf{r}, \mathbf{r}', t, t') = \theta(t - t')G_1(\mathbf{r}, \mathbf{r}', t, t')$$

and

$$G_-(\mathbf{r}, \mathbf{r}', t, t') = -\theta(t' - t)G_1(\mathbf{r}, \mathbf{r}', t, t') ,$$

depending on whether one uses $+i\epsilon$ or $-i\epsilon$ to avoid the singularity arising in the Fourier transform method for this case. In the above equation for G_+ and G_-, G_1 is given by Eq. (7.139) with $\tau = i\hbar t/2m$.

27. a) Show that the function

$$G_A(\mathbf{r}, \mathbf{r}', \tau, \tau') = -\theta(\tau' - \tau)\tilde{G}_2(\mathbf{r}, \mathbf{r}', \tau, \tau')$$

is indeed a Green's function for the operator

$$\theta = H + \frac{\partial^2}{\partial \tau^2}$$

if $\tilde{G}_2$ satisfies Eqs. (7.152) and (7.157).

b) For the case of the wave equation, find an expression for $G_A(\mathbf{r}, \mathbf{r}', \tau, \tau')$ analogous to the one for G_R given in Eq. (7.163).

c) Find a solution to $\theta\psi(\mathbf{r}, \tau) = J(\mathbf{r}, \tau)$ in terms of G_A. Interpret the solution by analogy with Eq. (7.164).

d) Show that

$$G_A + G_R = \frac{1}{2\pi}\delta(s^2) ,$$

where s^2 is the Lorentz-invariant scalar product:

$$s^2 \equiv (\mathbf{r}-\mathbf{r}')^2 - (\tau - \tau')^2 = (x - x')^2 + (y - y')^2 + (z - z')^2 - c^2(t - t')^2.$$

28. Using the Fourier transform method (in *four dimensions*) obtain the Green's functions for the wave equation. Show that one obtains G_R [of Eq. (7.163)] or G_A (of the previous problem) depending on how one chooses to avoid a singularity which arises in the Fourier transform method (see a similar discussion in Section 7.4, (Eq. (7.63) *et seq.*)

29. Consider the time-dependent inhomogeneous Klein-Gordon equation:

$$\left[-\nabla^2 + \frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \kappa^2\right]\psi(\mathbf{r}, t) = \rho(\mathbf{r}, t) .$$

a) Show that by using the Fourier transform method one obtains the following two integral expressions for the Green's functions G_R and G_A:

$$G_R(\mathbf{r}, t, \mathbf{r}', t') = \frac{c}{8\pi^2 Ri}\frac{d}{dR}\int_{-\infty}^{\infty} \frac{e^{i(R/c)\sqrt{q^2-\kappa^2c^2}}}{\sqrt{q^2 - \kappa^2c^2}} e^{-iq(t-t')}\, dq ,$$

$$G_A(\mathbf{r}, t, \mathbf{r}', t') = -\frac{c}{8\pi^2 Ri}\frac{d}{dR}\int_{-\infty}^{\infty} \frac{e^{-i(R/c)\sqrt{q^2-\kappa^2c^2}}}{\sqrt{q^2 - \kappa^2c^2}} e^{-iq(t-t')}\, dq ,$$

where $R = |\mathbf{r} - \mathbf{r}'|$. The two expressions correspond to two possible ways of avoiding a singularity which occurs in doing the Fourier transform integrals corresponding to the space variables. In the first integral, $\sqrt{q^2 - \kappa^2c^2}$ has *positive* imaginary part when $|q| < \kappa c$; in the second integral $\sqrt{q^2 - \kappa^2c^2}$ has *negative* imaginary part when $|q| < \kappa c$.

b) Show that $G_R = 0$ when $R/c > (t - t')$ and that $G_A = 0$ when $R/c + (t - t') > 0$. [*Hint*: If the integral for G_R is considered as an integral in the complex plane, then the plane is cut from $-\kappa c$ to $+\kappa c$ and the integral in question runs along the *top* of the cut. For G_A, the integral runs along the bottom of the cut.]

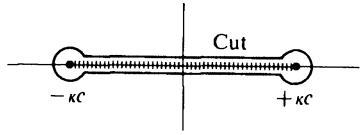

Fig. 7.12

c) When $R/c < (t - t')$, show that the integral occurring in G_R can be reduced to the contour integral shown in Fig. 7.12. Hence show that

$$G_R = -\frac{c}{4\pi^2 R}\frac{d}{dR}\left[\int_{-\kappa c}^{\kappa c}\frac{1}{\sqrt{\kappa^2c^2 - q^2}}\cosh\left[\frac{R}{c}\sqrt{\kappa^2c^2 - q^2}\right] \times e^{-iq(t-t')}\,dq\,\theta(T - R)\right],$$

and, similarly,

$$G_A = -\frac{c}{4\pi^2 R}\frac{d}{dR}\left[\int_{-\kappa c}^{\kappa c}\frac{1}{\sqrt{\kappa^2c^2 - q^2}}\cosh\left[\frac{R}{c}\sqrt{\kappa^2c^2 - q^2}\right] \times e^{-iq(t-t')}\,dq\,\theta(-T - R)\right],$$

where $\theta(x)$ equals 1 if $x > 0$ and equals zero if $x < 0$. Also, $T = c(t - t')$.

d) Making the change of variable $q = \kappa c \sin\theta$ and using some symmetry considerations, show that

$$G_R = -\frac{c}{8\pi^2 R}\frac{d}{dR}\left[\int_{-\pi}^{\pi}\cosh(\kappa R\cos\theta)e^{-i\kappa T\sin\theta}\,d\theta\,\theta(T - R)\right],$$

$$G_A = -\frac{c}{8\pi^2 R}\frac{d}{dR}\left[\int_{-\pi}^{\pi}\cosh(\kappa R\cos\theta)e^{-i\kappa T\sin\theta}\,d\theta\,\theta(-T - R)\right].$$

[*Hint*: Note that the limits of integration are $\pm\pi$, *not* $\pm\pi/2$.]

e) Show that this can be written as

$$G_R = -\frac{c}{8\pi^2 Ri}\frac{d}{dR}\left[\oint_C \frac{1}{z}\cosh\left[\frac{1}{2}\kappa R\left(z + \frac{1}{z}\right)\right]e^{-\kappa T(z-1/z)/2}dz\,\theta(T - R)\right],$$

with a similar expression for G_A. Here C is a counterclockwise contour around the *unit* circle.

f) By making two variable changes (in two different integrals), show that one obtains

$$G_R = -\frac{c}{8\pi^2 Ri}\frac{d}{dR}\left[\oint\frac{1}{\xi}\exp\left[-\frac{\kappa}{2}\sqrt{T^2 - R^2}\left(\xi - \frac{1}{\xi}\right)d\xi\,\theta(T - R)\right]\right].$$

Using the integral representation of the Bessel function [Eq. (6.65) and the sentence following this equation], show that this reduces to

$$G_R = -\frac{c}{4\pi R}\frac{d}{dR}[J_0(\kappa\sqrt{T^2 - R^2})\theta(T - R)].$$

Also,

$$G_A = -\frac{c}{4\pi R}\frac{d}{dR}[J_0(\kappa\sqrt{T^2 - R^2})\theta(-T - R)] .$$

[Here we have used the fact that $J_0(x) = J_0(-x)$, which is clear from Eq. (6.66)].

g) (We're almost done!) Carry out the differentiation using Eq. (7.22) to show that

$$G_R = \frac{c}{4\pi R}\delta(R - T) - \frac{\kappa c}{4\pi\sqrt{T^2 - R^2}}J_1(\kappa\sqrt{T^2 - R^2})\theta(T - R) ,$$

$$G_A = \frac{c}{4\pi R}\delta(R + T) - \frac{\kappa c}{4\pi\sqrt{T^2 - R^2}}J_1(\kappa\sqrt{T^2 - R^2})\theta(-T - R) .$$

Note that as $\kappa \to 0$, G_R goes over into the result of Eq. (7.163), apart from a factor of c which you should be able to explain (*Note*: $\delta(ax) = \delta(x)/|a|$). Also, note that $G_A + G_R$ can be written as

$$G_R + G_A = \frac{c}{2\pi}\delta(T^2 - R^2) - \frac{\kappa c}{4\pi\sqrt{T^2 - R^2}}J_1(\kappa\sqrt{T^2 - R^2})\theta(T^2 - R^2) ,$$

which is a function only of the Lorentz invariant $s^2 = R^2 - T^2 = (x - x')^2 + (y - y')^2 + (z - z')^2 - c^2(t - t')^2$.

FURTHER READINGS

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CHAPTER 8

INTRODUCTION TO INTEGRAL EQUATIONS

INTRODUCTION

In the last chapter, we saw that by introducing the concept of a Green's function we were led in a natural way to reformulate certain problems in terms of integral equations rather than differential equations. In this chapter and the next, we shall study integral equations as a subject in their own right, independent of the particular physical origin of these equations. We shall find that the relevant mathematical setting for this study is in infinite-dimensional vector spaces; within this framework, we shall concentrate primarily, although not exclusively, on Hilbert space. A particularly interesting question, in light of the work in Chapters 3 and 4, is: What is the relationship between the results of finite-dimensional and infinite-dimensional vector space theory? We will see that there is a very close relationship and that in many important cases one can solve infinite-dimensional problems by finite-dimensional techniques. However, there are also important and interesting differences which have many consequences for physics.

This chapter is devoted to a study of several special types of integral equations which can be solved rather easily. We will also take this opportunity to introduce, in a more precise manner than in Chapter 5, some of the key concepts in the theory of infinite-dimensional vector spaces.

8.1 ITERATIVE TECHNIQUES—LINEAR INTEGRAL OPERATORS

In Chapter 7 we encountered several different integral equations of the form

$$f(x) = g(x) + \int_a^b k(x, y) f(y)\, dy\,, \tag{8.1}$$

where $g(x)$ and $k(x, y)$ are given functions. Equations of this form are known as *integral equations of the second kind.* If $g(x)$ is not identically zero, the equation is said to be inhomogeneous; if $g(x)$ vanishes everywhere, the equation is called homogeneous. If on the left-hand side of Eq. (8.1) we replace $f(x)$ by zero, the resulting equation is an *integral equation of the first kind.* An example of such an equation is

$$g(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ixy} f(y)\, dy\,,$$

where $g(x)$ is a known function and $f(y)$ is to be determined. For this partic-

ular case, we know from the theory of Fourier transforms that the solution is given by

$$f(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iyx} g(x)\, dx\,,$$

but, in general, little is known about these equations and we will not discuss them further.

An important special case of Eq. (8.1) is the *Volterra* integral equation, which has the general form

$$f(x) = g(x) + \int_a^x \bar{k}(x, y) f(y)\, dy\,. \tag{8.2}$$

Equation (8.1) reduces to this form if the function $k(x, y)$ vanishes when $y > x$. A generalization of Eq. (8.2), which one might encounter, for example, in Newtonian mechanics, has the form

$$f(x) = g(x) + \int_a^x h(x, y, f(y))\, dy\,. \tag{8.3}$$

Here the unknown function f may occur linearly, as in Eq. (8.2), or nonlinearly, depending on the form of $h(x, y, f(y))$. In this chapter we will discuss primarily linear integral equations, although in Section 8.4 we will look briefly at some of the simpler aspects of nonlinear equations.

Let us now rewrite Eq. (8.1) in the form

$$f = g + Kf\,, \tag{8.4}$$

where K is an operator defined by

$$Kf \equiv \int_a^b k(x, y) f(y)\, dy\,. \tag{8.5}$$

Evidently K is linear, since

$$\begin{aligned} K(f_1 + f_2) &= \int_a^b k(x, y)[f_1(y) + f_2(y)]\, dy \\ &= \int_a^b k(x, y) f_1(y)\, dy + \int_a^b k(x, y) f_2(y)\, dy \\ &= Kf_1 + Kf_2\,. \end{aligned}$$

Therefore we may consider f and g as elements in some vector space, say V, and K as a linear operator which maps V into itself. The operator K, defined by Eq. (8.5), is called an *integral operator*, and $k(x, y)$ is called the *kernel* of the operator.

Equation (8.4) can now be stated as

$$(I - K)f = g\,, \tag{8.6}$$

where I is the identity operator. In this form the problem looks very much like the matrix equations discussed in Chapters 3 and 4. If we can compute $(I - K)^{-1}$, the problem is formally solved, since

$$f = (I - K)^{-1} g \tag{8.7}$$

(provided, of course, that the inverse operator is well defined). Note that although K may be an integral operator in Eq. (8.7), I is not an integral operator unless we want to use the δ-function:

$$If = \int_{-\infty}^{\infty} \delta(x - y) f(y)\, dy .$$

The most fortunate situation would be if K were in some sense a "small" operator, so that we could try to write

$$(I - K)^{-1} = I + K + K^2 + \cdots \tag{8.8}$$

by analogy with the power series for $(1 - x)^{-1}$ for small x. Since we have already required that K acting on *any* element of V produce another element of V, K^n is defined simply by successive applications of K: $K^2 = KK$, $K^3 = KK^2$, etc. This restriction on K is not trivial because it sometimes happens that reasonable-looking operators acting on V produce objects which are not in V. For example, consider the space $L_2[0, 1]$ of square-integrable functions of one variable defined on $[0, 1]$, and let the operator d/dx act on this space. Clearly, $f(x) = \sqrt{x}$ belongs to $L_2[0, 1]$, but

$$\frac{d}{dx}[f(x)] = \tfrac{1}{2}x^{-1/2}$$

does not. Thus d/dx does not transform every element of the space $L_2[0, 1]$ into another element of the same space, and cannot therefore be an admissible operator for our purposes. In fact, the assumption that a set of operators always transforms elements in a given space into elements of the *same* vector space has some strong consequences, as we shall point out later.

Suppose now that the right-hand side of Eq. (8.8) "converges." (We use quotation marks around "converges" because we have not yet given a careful definition of convergence for operators.) Then the operator to which it converges is the inverse of $(I - K)$, because multiplying $I + K + K^2 + K^3 + \cdots$ on either side by $(I - K)$ gives I. Thus we are led to speculate that the series

$$f = g + Kg + K^2g + K^3g + \cdots , \tag{8.9}$$

when it converges, is the solution to Eq. (8.4). The series in Eq. (8.9) is called the *Neumann series* by mathematicians, and the *Born series* by physicists, since it was Max Born who first utilized the basic iterative idea in quantum mechanics (although he was motivated by rather different considerations than the ones described here).

Let us now begin to investigate the circumstances under which the series of Eq. (8.9) converges. If K is an integral operator of the type defined by Eq. (8.5), then Eq. (8.9) can be written as

$$\begin{aligned} f(x) = g(x) &+ \int_a^b k(x, x')g(x')\, dx' \\ &+ \int_a^b dx' \int_a^b dx''\, k(x, x')k(x', x'')g(x'') + \cdots . \end{aligned} \tag{8.10}$$

We are, of course, assuming that $g(x) \not\equiv 0$; otherwise the series of Eq. (8.9) or (8.10) degenerates into the result $f(x) \equiv 0$, which is always a possible (but uninteresting) solution when $g(x) \equiv 0$. We require that $|k(x, y)|$ be *bounded* when x and y lie in $[a, b]$:

$$\max_{x, y \in a, b} |k(x, y)| = M, \tag{8.11}$$

where M is some finite number. We further assume that the integral

$$\int_a^b |g(x)|\, dx \tag{8.12}$$

exists and is equal to a finite constant C. Then we may write

$$f(x) = g_0(x) + g_1(x) + g_2(x) + g_3(x) + \cdots, \tag{8.13}$$

where $g_0(x) \equiv g(x)$ and

$$g_n(x) = \int_a^b k(x, y) g_{n-1}(y)\, dy.$$

We next want to show that, with the given assumptions about $k(x, y)$ and $g(x)$,

$$|g_n(x)| \leq CM[M(b - a)]^{n-1} \tag{8.14}$$

for $n \geq 1$. For $n = 1$,

$$|g_1(x)| \leq \int_a^b |k(x, y) g(y)|\, dy \leq \int_a^b |k(x, y)| \cdot |g(y)|\, dy.$$

But $|k(x, y)| \leq M$, so

$$|g_1(x)| \leq M \int_a^b |g(y)|\, dy = MC.$$

Thus Eq. (8.14) is satisfied for $n = 1$. We now assume it to be true for $n = m - 1$ and then prove that it is true for $n = m$. This will establish the general result by induction. For $g_m(x)$, we have

$$\begin{aligned} |g_m(x)| &\leq \int_a^b |k(x, y)| \cdot |g_{m-1}(y)|\, dy \leq MC[M(b - a)]^{m-2} \int_a^b |k(x, y)|\, dy \\ &\leq MC[M(b - a)]^{m-1}, \end{aligned}$$

which is the desired result. Thus the mth term in the series of Eq. (8.13) is *dominated* by $MC[M(b - a)]^{m-1}$, that is, it is less in absolute magnitude than $MC[M(b - a)]^{m-1}$, for all x. Therefore

$$\begin{aligned} \left| f(x) - \sum_{n=0}^{N} g_n(x) \right| &\leq |g_{N+1}(x)| + |g_{N+2}(x)| + \cdots \\ &\leq MC \sum_{m=N}^{\infty} [M(b - a)]^m \\ &\leq MC[M(b - a)]^N \sum_{m=0}^{\infty} [M(b - a)]^m. \end{aligned}$$

If $M(b-a) < 1$, this last sum converges, so we can write

$$|f(x) - \sum_{n=0}^{N} g_n(x)| \leq \frac{MC[M(b-a)]^N}{1 - M(b-a)}$$

for *all* x. Since $M(b-a) < 1$, the right-hand side of this equation can be made as small as we please by choosing N sufficiently large. Thus as N tends to infinity, $\sum_{n=0}^{N} g_n(x)$ converges uniformly to a function $f(x)$. The reader can easily show that if $k(x, y)$ is continuous in x for all y and if $g(x)$ is continuous, then each $g_n(x)$ will be continuous, and the uniformly convergent sum will also be continuous. Since the series $f = g + Kg + K^2g + \cdots$ is uniformly convergent, we can calculate $(I - K)f$ by integrating term by term. We obtain

$$(I - K)f = g_0 + g_1 + g_2 + g_3 + \cdots - K(g_0 + g_1 + g_2 + \cdots) .$$

But $Kg_n = g_{n+1}$; therefore

$$(I - K)f = g_0 \equiv g ,$$

so the Neumann series does indeed give a solution to the problem.

We may summarize by saying that if $M < (b-a)^{-1}$, then the Neumann series of Eqs. (8.9) or (8.10) converges to the solution of Eq. (8.1). This in no way implies that M *must* be less than $(b-a)^{-1}$ for the Neumann series to converge. It is easy to construct kernels for which $M \geq (b-a)^{-1}$ but for which the Neumann series nevertheless converges. For example, the reader should calculate the Neumann series for

$$k(x, y) = \xi(x)\eta(y) ,$$

where

$$\int_a^b \xi(x)\eta(x)\, dx = 0 .$$

Another large class of kernels for which the restriction $M < (b-a)^{-1}$ is unnecessary is the Volterra class, i.e., equations of the form of Eq. (8.2). For this case, if we write

$$f(x) = \bar{g}_0(x) + \bar{g}_1(x) + \bar{g}_2(x) + \cdots ,$$

where again $\bar{g}_0(x) \equiv g(x)$ and

$$\bar{g}_n(x) \equiv \int_a^x k(x, y)\bar{g}_{n-1}(y)\, dy ,$$

then we can show that

$$|\bar{g}_n(x)| \leq MC[M(x-a)]^{n-1}/(n-1)!$$

for $n \geq 1$, if $a \leq x \leq b$. Just as before, this is true for $n = 1$. Assuming it to be true for $n = m - 1$, we get

$$\begin{aligned} \bar{g}_m(x) &\leq \int_a^x |k(x, y)| \cdot |\bar{g}_{m-1}(y)|\, dy \\ &\leq MCM^{m-2} \int_a^x |k(x, y)|\, (y-a)^{m-2}\, dy/(m-2)! . \end{aligned}$$

Since $|k(x, y)| \leq M$, we have

$$|g_m(x)| \leq MC[M(x - a)]^{m-1}/(m - 1)!\,,$$

as required. Thus for $x \in [a, b]$, the nth term of the Neumann series is dominated by

$$MC[M(b - a)]^{n-1}/(n - 1)!\,,$$

so that the Neumann series is dominated by a series which converges for *all* values of $M(b - a)$. Hence the Neumann series *always* provides a solution to equations of the Volterra type.

8.2 NORMS OF OPERATORS

It may have occurred to the reader that the discussion of the previous section suffers from one serious limitation, namely, the requirement $M < (b - a)^{-1}$. In the case considered in Section 8.1, if the interval of integration is infinite, this requirement reduces to $M < 0$. But M is by definition positive, so no kernel can satisfy this requirement. However, to estimate a function by replacing it everywhere by its maximum value is a very crude approximation for infinite intervals. What we would like to be able to show is that we can replace

$$\max |k(x, y)| < (b - a)^{-1}$$

by a less restrictive requirement on $k(x, y)$. To do so we must first examine in some detail the problem of defining the magnitude of a linear *operator*.

In Hilbert space, we defined the norm of a *vector* f by

$$||f|| \equiv (f, f)^{1/2}\,,$$

where $(\ ,\ )$ is an inner product on the Hilbert space to which f belongs (see Definition 4.1). However, the concept of the norm of a vector is much more general; the existence of a norm is associated only with a definition of length, not of angle. There are many vector spaces on which one cannot define an inner product, but where nevertheless a length, or metric, can be defined. We can generalize the concept of norm to this wider class of spaces as follows.

Definition 8.1. A *norm* on a vector space, V, is a relation which assigns to any element, f, of V a real number denoted by $||f||$ such that

a) $||f + g|| \leq ||f|| + ||g||$, for all $f, g \in V$;
b) $||f|| = 0$ if and only if $f \equiv 0$;
c) $||af|| = |a| \cdot ||f||$ for any scalar a belonging to the field over which V is defined.

A vector space on which a norm can be defined is called a *normed vector space*. We have already studied one such space in some detail, namely, Hilbert space, where the norm of a vector can be derived from the existence of the inner product. Because of the axioms satisfied by the inner product (Definition 4.1), the norm defined in this way will have all the properties required by Definition 8.1.

Once a norm has been defined on a vector space, questions concerning the convergence of sequences of vectors can be considered.

Definition 8.2. A sequence of vectors $\{f_n\}$ in a normed vector space, V, is said to *converge in norm* to $f \in V$ if $||f_n - f|| \to 0$ as $n \to \infty$.

This generalizes to an arbitrary normed space the notion of mean convergence introduced in Section 5.2 for Hilbert space.

Definition 8.3. A sequence of vectors $\{f_n\}$ in a normed vector space V is called a *Cauchy sequence* if, given any $\epsilon > 0$, there exists a number N such that $||f_n - f_m|| < \epsilon$ whenever $m > N$ and $n > N$.

The most important property of a Cauchy sequence is that we need make no reference to a limit function. With this in mind, we make a final definition.

Definition 8.4. A normed vector space is said to be *complete* if every Cauchy sequence in the space tends to a vector which lies in the space.

A well-known vector space which is normed but not complete is the vector space of rational numbers with the usual Euclidean norm (absolute value). For example, the sequence $\{s_n\}$ where

$$s_n = \sum_{m=1}^{n} \frac{1}{m^2}$$

is a Cauchy sequence of rationals whose limit, $\pi^2/6$, is irrational.

Complete, normed vector spaces are called *Banach spaces*, after the Polish mathematician who played the key role in developing the theory of these spaces. A familiar Banach space is the space of continuous functions on the interval $[a, b]$, which we denote by $C[a, b]$. A norm for this space is

$$||f(x)|| = \max_{x \in [a, b]} |f(x)| .$$

By the triangle inequality for real numbers,

$$|f(x) + g(x)| \leq |f(x)| + |g(x)|$$

for any x. Thus, in particular,

$$\max_{x \in [a, b]} |f(x) + g(x)| \leq \max_{x \in [a, b]} |f(x)| + \max_{x \in [a, b]} |g(x)| ,$$

or

$$||f + g|| \leq ||f|| + ||g|| .$$

The other two requirements of Definition 8.1 are trivially satisfied. Clearly, $C[a, b]$ is a vector space, since the sum of two continuous functions is a continuous function. Finally, and most important, with a norm defined in this way $C[a, b]$ is a *complete* space, because the limit of a uniformly convergent sequence of continuous functions is continuous (see Problem 8.1). Thus $C[a, b]$ is a Banach space.

Another important family of Banach spaces are the so-called L_p spaces, defined to be the vector spaces of functions, $f(x)$, which satisfy

$$\int_D |f(x)|^p \, dx < \infty ,$$

where D is some subset (not necessarily finite) of the real line. A norm for this space is given by (see Problems 8.2 and 8.3)

$$||f|| = \left[\int_D |f(x)|^p \, dx\right]^{1/p}$$

and it can be shown (Generalized Riesz-Fisher theorem) that the space L_p is complete with this norm, and is thus a Banach space. We see, in particular, that the space, L_2, of square-integrable functions which we considered in Chapter 5 is a Banach space as well as being a Hilbert space. In fact, *any* Hilbert space is a Banach space since it is a complete inner-product space whose norm is automatically provided by the inner product.

Now that we have fixed the structure of Banach spaces, it is appropriate to consider how the space is changed by linear transformations. Throughout this chapter we shall limit the discussion to *bounded* linear transformations.

Definition 8.5. A transformation K on a normed linear vector space V is said to be a *bounded, linear transformation* if

a) $K(f + g) = Kf + Kg$ for all $f, g \in V$;
b) $K(af) = aKf$ for all $f \in V$ and all scalars a in the field over which V is defined;
c) there exists a constant M such that $||Kf|| \leq M||f||$ for all $f \in V$.

Definition 8.6. Let V be a normed vector space, and K a bounded transformation on V. The *smallest* M satisfying $||Kf|| \leq M||f||$ for all $f \in V$ is called the *norm* of K, which we will write as $||K||$. Note that we are using the same notation for the norm of an operator as for the norm of a function. To avoid confusion, vectors will always be written in lower case letters, operators in upper case letters.

It is not hard to see that $||K||$ as defined above is indeed a norm; for example

$$||(K_1 + K_2)f|| = ||K_1 f + K_2 f|| \leq ||K_1 f|| + ||K_2 f||$$

since $K_1 f$ and $K_2 f$ are elements of V. Applying Definition 8.6, we have, for all f,

$$\begin{aligned} ||(K_1 + K_2)f|| &\leq ||K_1|| \, ||f|| + ||K_2|| \, ||f|| \\ &\leq [||K_1|| + ||K_2||] \, ||f|| . \end{aligned}$$

But $||K_1 + K_2||$ is the smallest M satisfying $||(K_1 + K_2)f|| \leq M||f||$ for all f, so we have immediately

$$||K_1 + K_2|| \leq ||K_1|| + ||K_2|| .$$

The other properties of the norm of the operator follow similarly from the normed structure of the underlying vector space V. This leads to the conclusion that the set of all bounded linear transformations of a Banach space is a normed linear vector space—in fact, a Banach space, since the space can be shown to be complete. Now, what about the norm of the product of two bounded linear transformations on V?

$$||(K_1K_2)f|| = ||K_1(K_2f)|| \leq ||K_1||\,||K_2f|| \leq ||K_1||\,||K_2||\,||f||$$

for all f, so that $||K_1K_2|| \leq ||K_1||\,||K_2||$. In particular, this implies that

$$||K^n|| \leq ||K||^n . \tag{8.15}$$

The assumption that K is a bounded transformation has one particularly useful consequence. Consider a sequence of vectors, f_n, in V which converges in norm to some element $f \in V$, that is,

$$||f_n - f|| \to 0$$

as $n \to \infty$. It is natural to inquire whether or not the related sequence $\{Kf_n\}$ converges to Kf. To show that it does, consider the expression

$$||Kf_n - Kf|| = ||K(f_n - f)|| \leq ||K||\,||f_n - f|| ;$$

since $||K||$ is finite and $f_n \to f$ in norm, it follows that $Kf_n \to Kf$ in norm. This is called the *continuity property* of bounded linear transformations.

The introduction of an operator norm gives us a particularly powerful means of discussing the convergence of operator sequences.

Definition 8.7. We say that a sequence of operators, K_n, *converges in norm* to an operator, K, if $||K_n - K|| \to 0$ as $n \to \infty$.

There are, of course, other possible ways in which we could have defined the convergence of a sequence of operators. For example, we could say that $\{K_n\}$ converges to K if, for all $f \in B$,

$$||K_nf|| \to ||Kf|| . \tag{8.16}$$

This is certainly an acceptable way to define the convergence of operators, but it is *not* equivalent to convergence in norm.

To illustrate the difference between these two types of convergence in a Hilbert space, and also to introduce a particularly useful type of operator, we consider the so-called projection operators, P_n, which are defined for any orthonormal set of vectors, $\{\phi_i\}$, by

$$P_nf \equiv \sum_{i=1}^{n} (\phi_i, f)\phi_i . \tag{8.17}$$

We say that P_n is the projection operator onto the subspace of H spanned by the first n elements of the set $\{\phi_i\}$. It is clear from the definition of Eq. (8.17) that P_n is an Hermitian operator, since

$$(f, P_ng) = (P_nf, g) \tag{8.18}$$

for all $f, g \in H$. Also, P_n is idempotent:

$$P_n^2 = P_n . \tag{8.19}$$

(This last property is obvious geometrically, since after once projecting a function into a subspace of H, subsequent applications of the same projection leave the function unaffected.)

The operator norm of P_n satisfies

$$||P_n|| \leq 1 .$$

To show this, we consider

$$||P_n f|| = (P_n f, P_n f)^{1/2} = (f, P_n f)^{1/2} = [\sum_{i=1}^{n} (f, \phi_n)(\phi_n, f)]^{1/2} .$$

But by Bessel's inequality [Eq. (5.9)], we get immediately

$$||P_n f|| \leq ||f||$$

for all f. Hence $||P_n|| \leq 1$. The same proof may be applied to the operator $(I - P_n)$ to show that

$$||I - P_n|| \leq 1;$$

we now observe that both of the inequalities on the norms of P_n and $I - P_n$ can be converted into strict equalities by noting, for example, that there is always at least one f for which $||P_n f|| = ||f||$. In fact, whenever $f = \phi_1, \phi_2, \cdots, \phi_n$ we will have $||P_n f|| = ||f||$. Thus, $||P_n|| \geq 1$, and combining this with our previous result ($||P_n|| \leq 1$), we conclude that

$$||P_n|| = 1. \tag{8.20}$$

Similarly,

$$||I - P_n|| = 1. \tag{8.21}$$

The completeness of an orthonormal set $\{\phi_i\}$ as given in Definition 5.6 may be stated according to Eq. (5.10) in terms of projection operators as

$$\lim_{n\to\infty} (f, P_n f) = (f, f)$$

for all $f \in H$. Now using Eqs. (8.18) and (8.19), we have

$$||P_n f|| = (P_n f, P_n f)^{1/2} = (f, P_n^2 f)^{1/2} = (f, P_n f)^{1/2} ,$$

so

$$\lim_{n\to\infty} ||P_n f|| = (f, f)^{1/2} = ||If|| .$$

Thus P_n converges to I in the sense suggested by Eq. (8.16). Using a common terminology (albeit a somewhat misleading one), we may say that the sequence of operators $\{P_n\}$ *converges strongly* to I. If we think of the vectors as "points" in the vector space, then operators are analogous to functions; in this sense strong convergence of sequences of operators is analogous to the pointwise convergence of sequences of functions.

What about convergence in norm? For the sequence $\{P_n\}$ to converge to I in norm, given any ϵ, there must exist an N such that

$$||(P_n - I)f|| \leq \epsilon||f|| \tag{8.22}$$

whenever $n \geq N$, *for all* $f \in H$, that is, N is independent of f. Thus convergence in norm of operators is analogous to uniform convergence of a sequence of functions. But Eq. (8.22) cannot be satisfied, for, given any N, it is always possible to pick some function $h \in H$, in particular, $h = \phi_{N+1}$, such that

$$(P_N - I)h = (P_N - I)\phi_{N+1} = -\phi_{N+1} .$$

Then

$$||(P_N - I)h|| = ||h|| ,$$

so the sequence $\{P_n\}$ does *not* converge in norm to I. We see that convergence in norm is a *stronger* requirement than is strong convergence. In the sections that follow, when there can be no ambiguity, we shall often refer to convergence in norm simply as "convergence."

8.3 ITERATIVE TECHNIQUES IN A BANACH SPACE

Now that we have a method for specifying the magnitude of a linear operator, we can give a brief but complete account of the general iterative technique. Our discussion in this section will contain the results of Section 8.1 as a special case, but will also apply to infinite intervals and to operators more general than those of the integral type. We consider the operator equation

$$f = g + Kf$$

as an equation on a Banach space, B, with $g \in B$ and K a bounded linear transformation on B. We impose on the operator K, the restriction

$$||K|| < 1 , \tag{8.23}$$

which is analogous to the requirement $M(b - a) < 1$ of Section 8.1.

We want to show that the sequence of partial sums $\{f_n\}$, where

$$f_n = \sum_{m=0}^{n} K^m g ,$$

is a Cauchy sequence. Consider

$$||f_\nu - f_\mu|| = ||\sum_{m=\mu+1}^{\nu} K^m g|| \leq \sum_{m=\mu+1}^{\nu} ||K^m g|| \leq \sum_{m=\mu+1}^{\nu} ||K^m|| \, ||g|| ,$$

where we have taken $\nu > \mu$ for convenience. According to Eq. (8.15) $||K^m|| \leq ||K||^m$ so

$$||f_\nu - f_\mu|| \leq ||g|| \sum_{m=\mu+1}^{\nu} ||K||^m = ||g|| \, ||K||^{\mu+1} \sum_{m=0}^{\nu-\mu-1} ||K||^m .$$

But

$$\sum_{m=0}^{\nu-\mu-1} ||K||^m \leq \sum_{m=0}^{\infty} ||K||^m = (1 - ||K||)^{-1} ,$$

where we have summed the geometric series. Thus finally,

$$||f_\nu - f_\mu|| \leq ||g||\,||K||^{\mu+1}(1 - ||K||)^{-1}\,. \tag{8.24}$$

Since $||K|| < 1$, $(1 - ||K||)^{-1}$ is a finite number. Hence Eq. (8.24) tells us that it is possible to choose μ large enough so that $||f_\nu - f_\mu||$ is smaller than any preassigned number. Therefore $\{f_n\}$ is a Cauchy sequence, and, since the space B is complete (by definition),

$$f_n \to f\,,$$

where f is an element of B. Now, by the continuity property of bounded linear operators,

$$Kf_n \to Kf$$

in norm. But Kf_n is given by

$$Kf_n = K\sum_{m=0}^{n} K^m g = \sum_{m=1}^{n+1} K^m g = \sum_{m=0}^{n+1} K^m g - g = f_{n+1} - g\,.$$

Hence

$$(f_{n+1} - g) \to Kf$$

in norm. But $f_{n+1} \to f$ in norm, so

$$f = g + Kf\,.$$

Thus f is a solution of the basic equation. It is also the only solution, for suppose that

$$f_1 = g + Kf_1\,, \qquad f_2 = g + Kf_2\,.$$

Then

$$f_1 - f_2 = K(f_1 - f_2)\,,$$

and therefore

$$||f_1 - f_2|| = ||K(f_1 - f_2)|| \leq ||K||\cdot||f_1 - f_2||\,.$$

Since $||K|| < 1$, this means that $||f_1 - f_2|| = 0$, so the solution we have found is unique.

Thus when $||K|| < 1$ it is always possible to solve Eq. (8.4). It is worth emphasizing that this does not mean that for $||K|| \geq 1$ the Neumann series must diverge. We have only shown that it *necessarily* converges for $||K|| < 1$. Clearly, the smaller $||K||$, the more rapidly the series will converge. If $||K||$ is sufficiently small, even one or two terms of the Neumann series may be adequate for the purpose at hand. This is often the case in physical problems.

A particularly important application of these results is to the space L_2 of square-integrable functions. Consider the integral operator K defined by

$$Kf = \int k(x, y)\, f(y)\, dy\,,$$

acting on this space. If $k(x, y)$ is square-integrable as a function of two variables, so that

$$\int dx \int dy\, |k(x, y)|^2 < \infty\,, \tag{8.25}$$

then we can show that K is a transformation which maps every element of L_2 into an element of L_2. From Schwartz's inequality,

$$\left[\left|\int k(x, y)\, f(y)\, dy\right|\right]^2 \leq \int |k(x, y)|^2\, dy \cdot \int |f(y)|^2\, dy\,,$$

or

$$|Kf|^2 \leq \int |k(x, y)|^2\, dy \cdot ||f||^2\,.$$

Therefore

$$||Kf||^2 = \int |Kf|^2\, dx \leq ||f||^2 \int dx \int dy\, |k(x, y)|^2\,. \tag{8.26}$$

But this last integral is finite by Eq. (8.25), so $||Kf||$ is finite, as we asserted above.

Equation (8.26) also gives a useful estimate for the norm of K. From it we conclude that

$$||Kf|| \leq \left[\int dx \int dy\, |k(x, y)|^2\right]^{1/2} ||f||\,,$$

for all $f \in L_2$, and hence

$$||K|| \leq \left[\int dx \int dy\, |k(x, y)|^2\right]^{1/2}\,.$$

In particular, if

$$\int dx \int dy |k(x, y)|^2 < 1\,,$$

then $||K|| < 1$, and the above theory applies.

The integral form of Schrödinger's equation furnishes a good example of square-integrable kernels. According to Eqs. (7.57) and (7.68),

$$\psi(\mathbf{r}) = \xi(\mathbf{r}) + \int G_0(\mathbf{r}, \mathbf{r}', E)\, V(\mathbf{r}')\psi(\mathbf{r}')\, d\mathbf{r}'\,, \tag{8.27}$$

where

$$G_0(\mathbf{r}, \mathbf{r}', E) = -\frac{m}{2\pi h^2} \frac{e^{iq|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|}\,; \tag{8.28}$$

here $\xi(\mathbf{r}) \equiv 0$ for the bound state case ($E < 0$), and

$$\xi(\mathbf{r}) = \frac{A}{(2\pi)^{3/2}} e^{i\mathbf{q}\cdot\mathbf{r}}$$

for the scattering case ($E \geq 0$). Also, q is defined by

$$q = |\mathbf{q}| = \sqrt{2mE/\hbar^2} ,$$

and when $E < 0$ we take q with *positive* imaginary part. It will be convenient in what follows to consider q to be a complex number with positive imaginary part. Now the kernel of the above equation is

$$k(\mathbf{r}, \mathbf{r}', E) = G_0(\mathbf{r}, \mathbf{r}', E)\, V(\mathbf{r}') ,$$

so

$$||K||^2 \leq \int d\mathbf{r} \int d\mathbf{r}' \, |k(\mathbf{r}, \mathbf{r}', E)|^2 = \int d\mathbf{r} \int d\mathbf{r}' \, |G_0(\mathbf{r}, \mathbf{r}', E)\, V(\mathbf{r}')|^2 .$$

Using Eq. (8.28) for $G_0(\mathbf{r}, \mathbf{r}', E)$ and writing $q = \operatorname{Re} q + i \operatorname{Im} q$, we get

$$||K||^2 \leq \left(\frac{m}{2\pi\hbar^2}\right)^2 \int d\mathbf{r} \int d\mathbf{r}' \, \frac{e^{-2 \operatorname{Im} q |\mathbf{r} - \mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|^2} \, |V(\mathbf{r}')|^2 .$$

If we change variables by setting $\mathbf{r} = \boldsymbol{\rho} + \mathbf{r}'$, the integral can be handled easily. The result is

$$||K||^2 \leq \frac{m^2}{2\pi\hbar^4 \operatorname{Im} q} \int |V(\mathbf{r})|^2 \, d\mathbf{r} , \tag{8.29}$$

which is finite *except* when $\operatorname{Im} q = 0$. But unfortunately this corresponds to the most interesting case, $E > 0$. This is when we are in the scattering region and would hope to be able to iterate Schrödinger's integral equation, at least under some conditions.

The reason that we encounter this difficulty is simply that we have failed to take adequate account of the physical situation. If $E > 0$, we are really interested in obtaining the scattering amplitude, $f(\theta, \phi)$, as was emphasized in Section 7.4. In terms of the wave function, $\psi(\mathbf{r})$, of Eq. (8.27),

$$f(\theta, \phi) = -\frac{m}{2\pi\hbar^2} \int e^{-i\mathbf{k}_f \cdot \mathbf{r}} V(\mathbf{r}) \psi(\mathbf{r}) \, d\mathbf{r} .$$

From this expression, it is clear that we need only know $\psi(\mathbf{r})$ in those regions of space where the magnitude (strength) of $V(\mathbf{r})$ is significant, i.e., three-dimensional space should be weighted in importance by the function $V(\mathbf{r})$. With this in mind, we can write the above expression for $f(\theta, \phi)$ as

$$f(\theta, \phi) = -\frac{m}{2\pi\hbar^2} (\psi_{\text{free}}, \psi)_\rho ,$$

where $\psi_{\text{free}} = e^{i\mathbf{k}_f \cdot \mathbf{r}}$ and the inner product $(\,,\,)_\rho$ is taken with respect to the weight function $V(\mathbf{r})$:

$$(f, g)_\rho \equiv \int f^*(\mathbf{r}) g(\mathbf{r}) \, d\rho\,(\mathbf{r}) , \tag{8.30a}$$

where $\rho(\mathbf{r})$ is defined by

$$d\rho\,(\mathbf{r}) \equiv V(\mathbf{r}) r^2 \sin\theta \, dr \, d\theta \, d\phi . \tag{8.30b}$$

Clearly, for this to be an inner product in the sense of Definition 4.1, $V(\mathbf{r})$ must not change sign as $\mathbf{r}$ varies. We will assume that this is the case; if it is not, the necessary modifications should be obvious to the reader. In this notation we may rewrite Eq. (8.27) as

$$\psi(\mathbf{r}) = \xi(\mathbf{r}) + \int G_0(\mathbf{r}, \mathbf{r}', E)\psi(\mathbf{r}')\, d\rho\,(\mathbf{r}') \,, \tag{8.31}$$

which we would like to consider as an integral equation on the Hilbert space whose inner product is given by Eq. (8.30a). This will be legitimate if $||\xi|| < \infty$, which will be the case if

$$\int d\rho\,(\mathbf{r}) = \int V(\mathbf{r})d\mathbf{r} < \infty \,,$$

since $|\xi(\mathbf{r})| = |Ae^{i\mathbf{k}_i\cdot\mathbf{r}}|$ is a constant. For most reasonable potentials $||\xi||$ is finite, although for the Coulomb potential $||\xi||$ is infinite. To be more precise, $||\xi||$ will be finite if $V(\mathbf{r})$ falls off faster than $1/r^3$ at large distances and is less singular than $1/r^3$ at the origin. For Eq. (8.31) to be an acceptable equation in Hilbert space, the norm of the integral operator in Eq. (8.31) must be finite. Denoting the integral operator whose kernel is $G_0(\mathbf{r}, \mathbf{r}', E)$ by K, we have

$$\begin{aligned} ||K||^2 &\leq \int d\rho\,(\mathbf{r}) \int d\rho\,(\mathbf{r}')\, |G_0(\mathbf{r}, \mathbf{r}', E)|^2 \\ &= \left(\frac{m}{2\pi\hbar^2}\right)^2 \int d\mathbf{r} \int d\mathbf{r}'\, V(\mathbf{r})\, V(\mathbf{r}')\, \frac{e^{-2\,\mathrm{Im}\,q|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|^2} \,. \end{aligned}$$

It is a relatively simple matter to evaluate this integral for the *central* potentials which typically occur in practice. The reason for this is that the integrand then depends only on $|\mathbf{r}|$, $|\mathbf{r}'|$, and $|\mathbf{r} - \mathbf{r}'|$, which form sides of a triangle. The integration over the angles which give the orientation of the triangle in space gives just $8\pi^2$. Writing $s = r + r'$, $t = r - r'$ and $u = |\mathbf{r} - \mathbf{r}'|$, one can easily show that for kernels which depend only on s, t, and u the volume element $d\mathbf{r}\, d\mathbf{r}'$ reduces to just $\pi^2(s^2 - t^2)u\, ds\, dt\, du$. The integration limits are defined by the inequalities $-u < t < u < s$. Thus

$$||K||^2 \leq \left(\frac{m}{2\hbar^2}\right)^2 \int_0^\infty ds \int_0^s du \int_{-u}^u dt\, (s^2 - t^2)\, \frac{e^{-2u\,\mathrm{Im}\,q}}{u}\, V(r)\, V(r') \,.$$

If, for example, $V(r)$ is the Yukawa potential, $V(r) = g^2e^{-\mu r}/r$, then since $s = r + r'$ and $rr' = (s^2 - t^2)/4$, we get

$$||K||^2 \leq \frac{m^2g^4}{\hbar^4} \int_0^\infty ds\, e^{-\mu s} \int_0^s \frac{du}{u}\, e^{-2u\,\mathrm{Im}\,q} \int_{-u}^u dt \,.$$

Doing the integrations, we find that

$$||K|| \leq \left[\frac{2}{1 + 2\,\mathrm{Im}\,(q/\mu)}\right]^{1/2} \frac{mg^2}{\mu\hbar^2} \,. \tag{8.32a}$$

Thus when $E > 0$ (that is, $\mathrm{Im}\,q = 0$),

$$||K|| \leq \sqrt{2}\,\frac{mg^2}{\mu\hbar^2} \,. \tag{8.32b}$$

Hence if the potential is weak enough, i.e., if g is small enough so $||K|| < 1$, we can iterate Schrödinger's integral equation. The iteration will converge in the weighted norm provided by Eq. (8.30a).

In the interest of not deceiving the reader, it should be emphasized that although the iteration is certain to converge if $V(\mathbf{r})$ is not too strong (and not too singular), it is, nevertheless, extremely difficult to perform the integrations involved in computing the successive terms of the series. If the reader has any doubts on this score, he is invited to try evaluating the first term in the iteration of Eq. (8.27) when $V(\mathbf{r})$ is given by some simple potential like the Yukawa potential. As we shall see, there are other methods of solution available which are often easier to employ than the method presented here. However, the Neumann series is a very important theoretical tool for developing other more practical methods of solution.

8.4 ITERATIVE TECHNIQUES FOR NONLINEAR EQUATIONS

Before leaving the subject of iterative solutions of integral equations, it is worthwhile contrasting the simple situation which prevails in the linear case with the more complicated features which arise in the nonlinear case. We shall see that the power of the iterative procedure is sharply reduced by nonlinearity, even when we restrict our attention to equations analogous to the Volterra equation discussed in Section 8.1.

Let us consider an equation of the form

$$f(x) = g(x) + \int_a^x h(x, y, f(y))\, dy\,, \qquad a \leq x \leq b\,. \tag{8.33}$$

We would like to develop an iterative procedure similar to the one we just employed to handle the linear case. However, since, in general, $f(y)$ does not occur linearly under the integration sign in Eq. (8.33), we shall proceed as follows. Set $f_0(x) \equiv g(x)$ and then by induction define

$$f_n(x) \equiv g(x) + \int_a^x h(x, y, f_{n-1}(y))\, dy\,. \tag{8.34}$$

The reader can easily verify that in the linear case, that is, $h(x, y, f(y)) = k(x, y) f(y)$, $f_n(x)$ is given by

$$f_n(x) = \sum_{m=0}^{n} K^m g\,,$$

where K is defined by Eq. (8.5). With this in mind, the basic idea is very simple. We start with $f_0(x) \equiv g(x)$ as a zeroth-order approximation to $f(x)$ and insert this approximation into Eq. (8.34) to obtain a first approximation, $f_1(x)$. Continuing this process, we would hope to generate a sequence which converges eventually to the solution to Eq. (8.33). One might expect that just as in the case of the Volterra linear integral equation, this process will always converge if $h(x, y, f(y))$ is bounded in the domain $a \leq x \leq b$, $a \leq y \leq b$. However, we must also remember that h depends on a third variable, namely, f itself.

Therefore we shall make the additional assumption that for (x, y) in the square $a \leq x \leq b$, $a \leq y \leq b$,

$$|h(x, y, f(y))| \leq M \tag{8.35}$$

whenever

$$|f(y) - g(y)| \leq \Delta , \tag{8.36}$$

where Δ is some given positive number and $y \in [a, b]$. These assumptions will at least enable us to start the iterative procedure.

Setting $f_0(x) = g(x)$, we obtain

$$\begin{aligned} f_1(x) &\equiv g(x) + \int_a^x h(x, y, f_0(y))\, dy \\ &= g(x) + \int_a^x h(x, y, g(y))\, dy , \end{aligned}$$

from which we may deduce that

$$|f_1(x) - g(x)| = |f_1(x) - f_0(x)| \leq M(x - a) , \tag{8.37}$$

since $|f_0(y) - g(y)| = 0 < \Delta$. But to continue one step further, we need to know that $|f_1(y) - g(y)| \leq \Delta$. Equation (8.37) tells us that this will be the case if $|y - a| \leq \Delta/M$. Thus, to get any further, we must restrict the variables so that

$$a \leq x \leq (\Delta/M) + a , \qquad a \leq y \leq (\Delta/M) + a . \tag{8.38}$$

Clearly, if $(\Delta/M) + a \geq b$, this is no restriction at all. Thus we can calculate

$$f_2(x) \equiv g(x) + \int_a^x h(x, y, f_1(y))\, dy .$$

Since $a \leq x \leq (\Delta/M) + a$, we have $|f_1(x) - g(x)| \leq \Delta$, and

$$|f_2(x) - g(x)| \leq M(x - a) \leq \Delta ;$$

this will be true for all further iterations. However, we also hope to show, in analogy with the linear case, that f_n and f_{n-1} get very close to each other when n becomes sufficiently large. Hence we need to calculate the difference between successive approximations. Let us look first at $|f_2(x) - f_1(x)|$. We find that

$$|f_2(x) - f_1(x)| = \left| \int_a^x [h(x, y, f_1(y)) - h(x, y, f_0(y))]\, dy \right| . \tag{8.39}$$

For linear integral equations, it is relatively easy to estimate the difference occurring under the integral in Eq. (8.39), because

$$h(x, y, f(y)) = k(x, y) f(y) .$$

But in the nonlinear case it is necessary to put further restrictions on the function h in the region defined by Eq. (8.38). By analogy with the linear case, we require that there exist some finite number, N, such that

$$|h(x, y, \phi(y)) - h(x, y, \psi(y))| \leq N|\phi(y) - \psi(y)| \tag{8.40}$$

whenever $|\phi(y) - g(y)| \leq \Delta$ and $|\psi(y) - g(y)| \leq \Delta$. This requirement is called the *Lipschitz condition*.

Returning now to Eq. (8.39), we see that since $|f_1(y) - g(y)| \leq \Delta$ and $|f_0(y) - g(y)| = 0 < \Delta$, we can use the Lipschitz condition to get

$$|f_2(x) - f_1(x)| \leq N \int_a^x |f_1(y) - f_0(y)|\, dy .$$

But $|f_1(y) - f_0(y)| = |f_1(y) - g(y)| \leq \Delta$, so

$$|f_2(x) - f_1(x)| \leq N\Delta(x - a) .$$

Continuing to the next step,

$$|f_3(x) - f_2(x)| \leq \Delta N^2(x - a)^2/2! ,$$

since $|f_2(x) - g(x)| \leq \Delta$. In general,

$$|f_n(x) - g(x)| \leq \Delta \tag{8.41}$$

and

$$|f_n(x) - f_{n-1}(x)| \leq \Delta N^{n-1}(x - a)^{n-1}/(n - 1)! . \tag{8.42}$$

In view of Eq. (8.38), Eq. (8.42) can be written as

$$|f_n(x) - f_{n-1}(x)| \leq \Delta(\Delta N/M)^{n-1}/(n - 1)! . \tag{8.43}$$

Thus for all $x \in [a, (\Delta/M) + a]$, $|f_n(x) - f_{n-1}(x)| \to 0$ uniformly as $n \to \infty$.

Just as in the linear case, we expect that $f_n(x)$ should tend to the solution of Eq. (8.33). To see this, we write $f_\nu(x)$ in the form

$$f_\nu(x) = f_0(x) + \sum_{n=1}^{\nu} [f_n(x) - f_{n-1}(x)] .$$

According to Eq. (8.43), each term of this sum is dominated by

$$\Delta(\Delta N/M)^{n-1}/(n - 1)! \quad \text{for} \quad x \in [a, (\Delta/M) + a] .$$

Thus as ν goes to infinity, $f_\nu(x)$ tends uniformly toward some function, say, $f(x)$. We now show that $f(x)$ satisfies Eq. (8.33). We write

$$\begin{aligned} f_\nu(x) &= g(x) + \int_a^x h(x, y, f_{\nu-1}(y))\, dy \\ &= g(x) + \int_a^x h(x, y, f(y))\, dy + \int_a^x [h(x, y, f_{\nu-1}(y)) - h(x, y, f(y))]\, dy . \end{aligned}$$

Taking the limit $\nu \to \infty$ on both sides of this equation, we obtain

$$f(x) = g(x) + \int_a^x h(x, y, f(y))\, dy + \lim_{\nu\to\infty} R_\nu ,$$

where

$$R_\nu = \int_a^x [h(x, y, f_{\nu-1}(y)) - h(x, y, f(y))]\, dy .$$

Since $|f(y) - g(y)| < \Delta$ because of Eqs. (8.41) and (8.43), we can use the Lipschitz condition to get

$$\begin{aligned} |R_\nu| &\leq N \int_a^x |f_{\nu-1}(y) - f(y)|\, dy \\ &\leq N\Delta \sum_{n=\nu}^{\infty} \frac{N^{n-1}}{(n-1)!} \int_a^x (y-a)^{n-1}\, dy \\ &\leq \Delta \sum_{n=\nu}^{\infty} \frac{N^n (x-a)^n}{n!} \leq \Delta \sum_{n=\nu}^{\infty} \left(\frac{N\Delta}{M}\right)^n \frac{1}{n!}\,, \end{aligned}$$

which tends to zero as $\nu \to \infty$. Thus

$$f(x) = g(x) + \int_a^x h(x, y, f(y))\, dy\,,$$

and $f(x)$ is indeed the desired solution to Eq. (8.33).

We have therefore obtained an iterative solution in an interval, the length of which will generally depend on specific properties of $h(x, y, f(y))$. In the linear Volterra case, we were able to exhibit a convergent iterative series for *all* values of x, so we see that nonlinearity has a very serious effect on the iterative technique. The appropriate *region* of convergence is controlled by Δ/M, while the *rate* of convergence depends on the size of N, the constant in the Lipschitz condition [Eq. (8.40)].

As an example of nonlinear perturbation theory, we consider a problem from classical mechanics. Suppose that a particle is bound by a slightly anharmonic spring, with potential energy

$$V(x) = \tfrac{1}{2}kx^2 + \tfrac{1}{3}ax^3\,,$$

where a is a small constant. The equation of motion is

$$m\ddot{x} + kx + ax^2 = 0\,, \qquad \text{or} \qquad \ddot{x} + \omega_0^2 x = -\epsilon x^2\,,$$

where $\omega_0^2 = k/m$ and $\epsilon = a/m > 0$.

According to Eq. (7.35) (with $\gamma = 0$) the Green's function for the operator $d^2/dt^2 + \omega_0^2$ with one-point boundary conditions is

$$G(t, t') = \frac{1}{\omega_0} \sin\,[\omega_0(t - t')]\theta(t - t')\,.$$

Assuming that the boundary condition is applied at $t = 0$ and has the form $x(0) = x_0 > 0$, $\dot{x}(0) = 0$, we get

$$x(t) = x_0 \cos \omega_0 t - \frac{\epsilon}{\omega_0} \int_0^t \sin \omega_0(t - t') x^2(t')\, dt'\,.$$

In keeping with the previous discussion, let us inquire under what conditions we can obtain an iterative solution for $x(t)$ if we require that

$$|x(t) - x_0 \cos \omega_0 t| < x_0\,. \tag{8.44}$$

For this particular problem, we have also

$$h(t, t', x(t')) = -(\epsilon/\omega_0) \sin \omega_0(t - t')x^2(t') .$$

Thus

$$|h(t, t', x(t'))| \leq (\epsilon/\omega_0)x^2(t') ,$$

which may be written as

$$\begin{aligned} |h(t, t', x(t'))| &\leq (\epsilon/\omega_0)[x(t') - x_0 \cos \omega_0 t' + x_0 \cos \omega_0 t']^2 \\ &\leq (\epsilon/\omega_0)[|x(t') - x_0 \cos \omega_0 t'| + x_0|\cos \omega_0 t'|]^2 . \end{aligned}$$

Using Eq. (8.44), we obtain immediately

$$|h(t, t', x(t'))| < 4\epsilon x_0^2/\omega_0 , \tag{8.45}$$

for all t and t' whenever $|x(t') - x_0 \cos \omega_0 t'| \leq x_0$.

Similarly, we may estimate the quantity

$$|h(t, t', x(t')) - h(t, t', \xi(t'))|$$

which appears in the Lipschitz condition. We have

$$|h(t, t', x(t')) - h(t, t', \xi(t'))| \leq (\epsilon/\omega_0)|x(t') + \xi(t')|\,|x(t') - \xi(t')| ,$$

which may be rewritten as

$$\begin{aligned} &|h(t, t', x(t')) - h(t, t', \xi(t'))| \\ &\quad \leq (\epsilon/\omega_0)|x(t') - x_0 \cos \omega_0 t' + \xi(t') - x_0 \cos \omega_0 t' + 2x_0 \cos \omega_0 t'| \cdot \\ &\qquad\qquad |x(t') - \xi(t')| . \end{aligned}$$

Using Eq. (8.44) again, we have

$$|h(t, t', x(t')) - h(t, t', \xi(t'))| \leq (4\epsilon x_0/\omega_0)|x(t') - \xi(t')| \tag{8.46}$$

for all t and t' whenever $|x(t') - x_0 \cos \omega_0 t'| \leq x_0$ and $|\xi(t') - x_0 \cos \omega_0 t'| \leq x_0$. Thus the quantities Δ, M, and N are given by

$$\Delta = x_0 , \tag{8.47a}$$

$$M = 4\epsilon x_0^2/\omega_0 , \tag{8.47b}$$

$$N = 4\epsilon x_0/\omega_0 . \tag{8.47c}$$

The above estimates involve no restrictions on t, but the previous discussion of the iterative method in the nonlinear case tells us that the iterative solution will be valid only in the interval $0 \leq t \leq \Delta/M$ or $0 \leq \omega_0 t \leq 1/(4\delta)$, where $\delta = \epsilon x_0/\omega_0^2$, and δ is the parameter in which we are attempting to expand $x(t)$. Hence we see that if δ is small, the expansion will converge for a large range of times.

With these results in hand, we begin the iterative procedure by setting

$$x_0(t) = x_0 \cos \omega_0 t . \tag{8.48}$$

By our previous work,

$$x_1(t) = x_0 \cos \omega_0 t - (\epsilon/\omega_0) \int_0^t \sin \omega_0(t - t')x_0^2(t')\, dt' .$$

Using Eq. (8.48) for $x_0(t')$, we get

$$x_1(t) = x_0\,[\cos\omega_0 t + \delta(\tfrac{1}{6}\cos 2\omega_0 t + \tfrac{1}{3}\cos\omega_0 t - \tfrac{1}{2})]\,. \tag{8.49}$$

Note that $x_1(t)$ satisfies the boundary conditions at $t = 0$.

The main effect of the perturbation in first approximation is a slight change in the amplitude of the dominant harmonic ($\cos\omega_0 t$) and the addition of a second harmonic term ($\cos 2\omega_0 t$) with small amplitude (assuming that δ is small). By looking at $x_1(t)$ alone, it is not clear why the restriction $0 \leq \omega_0 t \leq 1/(4\delta)$ is necessary, since the size of the first-order correction is on the average independent of t. To see the reason for this restriction we look at $x_2(t)$. A few integrations show that

$$\begin{aligned} x_2(t) = x_0[&\cos\omega_0 t + \delta(\tfrac{1}{6}\cos 2\omega_0 t + \tfrac{1}{3}\cos\omega_0 t - \tfrac{1}{2}) \\ &+ \delta^2(\tfrac{1}{48}\cos 3\omega_0 t + \tfrac{1}{9}\cos 2\omega_0 t + \tfrac{29}{144}\cos\omega_0 t - \tfrac{1}{3}) \\ &+ \delta^3(\tfrac{1}{1080}\cos 4\omega_0 t + \tfrac{1}{144}\cos 3\omega_0 t - \tfrac{1}{27}\cos 2\omega_0 t + \tfrac{753}{2160}\cos\omega_0 t - \tfrac{23}{72}) \\ &+ \omega_0 t(\tfrac{5}{12}\delta^2\sin\omega_0 t + \tfrac{5}{36}\delta^3\sin\omega_0 t)]\,. \end{aligned} \tag{8.50}$$

This iterate has the same general form as $x_1(t)$ expect for the appearance of terms linear in $\omega_0 t$. Thus for t large, both $|x_2(t) - x_1(t)|$ and $|x_2(t) - x_0(t)|$ are dominated by terms proportional to t. These terms will remain bounded only if we restrict the range of t.

It should be noted that $x_2(t)$, although it contains terms of order δ^3, is complete only through terms of order δ^2; there are further terms of order δ^3 which will make their appearance in $x_3(t)$, along with terms of still higher order. (In general, there will be terms proportional to $\delta^{(2^n-1)}$ in the nth iterate.) It is also interesting to note that if we define

$$\omega_1 \equiv \omega_0[1 - \tfrac{5}{6}\delta^2]^{1/2}\,, \tag{8.51}$$

and replace ω_0 by ω_1 in $\cos\omega_0 t$, then through order δ^2, $x_2(t)$ can be written as in Eq. (8.50), but with the omission of the term linear in t and proportional to δ^2. This follows from the fact that through order δ^2

$$\cos\omega_0 t + \tfrac{5}{12}\delta^2\omega_0 t\sin\omega_0 t \simeq \cos\omega_1 t\,,$$

where ω_1 is given by Eq. (8.51). Thus, by redefining the frequency, it is possible to eliminate terms proportional to t, at least through order δ^2. It can be shown (see Problem 8.15) that a similar elimination can be achieved for all terms through order δ^3.

8.5 SEPARABLE KERNELS

We have now seen in some detail how to solve linear equations involving operators of small norm. However, it is clear, particularly from the example discussed at the end of Section 8.3, that there are many interesting problems which involve operators of finite, but not necessarily small, norm. We must therefore develop techniques of more general applicability.

We might hope that it will prove possible to treat operators on infinite-dimensional spaces in the same way that we treated operators on finite-dimen-

sional spaces (Chapters 3 and 4). There we could simply take an operator, A, and turn it into a finite-dimensional matrix by computing its inner product between the finite number of vectors which constitute an orthonormal basis for the space. However, on infinite-dimensional spaces there must be an infinite number of vectors in any basis, so this procedure would lead us to infinite-dimensional matrices, and we have said nothing about such objects in previous chapters. (Even the problem of multiplying two infinite-dimensional matrices is not a trivial one because of convergence questions.)

On the other hand, we know that if we write a function, f, in terms of an orthonormal basis, $\{\phi_n\}$, of Hilbert space by computing the expansion coefficients, $a_n = (\phi_n, f)$, we can associate the infinite collection of a_n with an infinite column vector, and, more important, as n becomes large, say $n > N$, the elements a_n with $n > N$ become *small* (Bessel's inequality). Thus any function in Hilbert space can be approximated arbitrarily closely by a *finite* linear combination,

$$f \equiv \sum_{n=1}^{N} a_n \phi_n .$$

This encourages us to believe that perhaps the same situation will prevail for linear operators; namely, if we start computing the matrix elements of an operator, A, between the elements of an infinite orthonormal basis, then after we have computed enough matrix elements, A_{mn}, the elements will start to become small, just as did the a_n in the case of vectors. Thus we might hope that we can stop completely when we get to a large $N \times N$ array and just forget about the elements A_{mn}, with m or n greater than N. The value of N would, of course, have to depend on the degree of precision which one wanted in a given problem, but we imagine that any degree of precision can be obtained by this method.

If this were possible, the problem would come down to simple matrix diagonalization or inversion, which we can carry out by employing the methods of Chapters 3 and 4. We shall see that there is a rather large class of operators for which such an approach will work. It is the purpose of the subsequent sections of this chapter and also of the following chapter to present in some detail the theory of such operators.

It is best, however, to start with a type of operator for which no approximation is necessary in the reduction to a simple matrix. This is the class of operators of *finite rank*, so called because they are directly reducible to finite-dimensional matrices (and hence, by the discussion in Section 3.10, to matrices of finite rank).

To illustrate what we have in mind, let us consider a separable kernel, i.e., one which has the form

$$k_s(x, y) = \lambda \phi(x) \psi^*(y) , \tag{8.52}$$

where for the sake of simplicity we shall assume that ϕ and ψ are functions in Hilbert space. If

$$\int |k_s(x, y)|^2 \, dx \, dy < 1 , \tag{8.53}$$

then we know from our work in the previous section that the Neumann series converges. Let us look at this series. The iterative solution to the equation

$$f(x) = g(x) + \int k_s(x, y) f(y)\, dy \tag{8.54}$$

is

$$f(x) = g(x) + \int dy_1\, k_s(x, y_1)\, g(y_1) + \int dy_1 \int dy_2\, k_s(x, y_1) k_s(y_1, y_2)\, g(y_2) + \cdots$$

Since $k_s(x, y)$ is square integrable as a function of two variables, $\phi(x)$ and $\psi(y)$ must be square integrable as functions of a single variable. Thus if $g(x)$ is square integrable, then all the integrals in the iteration are well defined, and thus

$$\begin{aligned} f(x) = g(x) &+ \lambda \int dy_1\, \phi(x) \psi^*(y_1) g(y_1) \\ &+ \lambda^2 \int dy_1 \int dy_2\, \phi(x) \psi^*(y_1) \phi(y_1) \psi^*(y_2) g(y_2) + \cdots \end{aligned}$$

Let us set

$$\int \psi^*(y) \phi(y)\, dy = (\psi, \phi)\ .$$

Then $f(x)$ can be written as

$$\begin{aligned} f(x) &= g(x) + \lambda \phi(x)(\psi, g) + \lambda^2 \phi(x)(\psi, \phi)(\psi, g) \\ &\quad + \lambda^3 \phi(x)(\psi, \phi)^2 (\psi, g) + \cdots \\ &= g(x) + \lambda \phi(x)(\psi, g) \sum_{n=0}^{\infty} \lambda^n (\psi, \phi)^n\ . \end{aligned} \tag{8.55}$$

If $|\lambda(\psi, \phi)| < 1$, the series of Eq. (8.55) converges. In fact, $|\lambda(\psi, \phi)|$ *is* less than unity. To prove this, note that Eq. (8.53) implies

$$|\lambda|^2 \int |\phi(x)|^2\, dx \int |\psi(y)|^2\, dy < 1\ . \tag{8.56}$$

But by Schwartz's inequality,

$$|(\psi, \phi)| \leq [(\phi, \phi)(\psi, \psi)]^{1/2}\ .$$

Thus, by Eq. (8.56),

$$|\lambda(\psi, \phi)| \leq [|\lambda|^2 (\phi, \phi)(\psi, \psi)]^{1/2} < 1\ ,$$

and we can sum the geometric series in Eq. (8.55) directly:

$$f(x) = g(x) + \lambda \frac{(\psi, g)}{1 - \lambda(\psi, \phi)} \phi(x)\ . \tag{8.57}$$

Although we derived this simple result in the framework of the Neumann series, its closed form suggests that it might be more general. That this is indeed the case may be seen by writing Eq. (8.54) as

$$\begin{aligned} f(x) &= g(x) + \lambda \int \phi(x) \psi^*(y) f(y)\, dy \\ &= g(x) + \lambda \phi(x)(\psi, f)\ . \end{aligned} \tag{8.58}$$

If we take the inner product of both sides of Eq. (8.58) with ψ, we get

$$(\psi, f) = (\psi, g) + \lambda(\psi, \phi)(\psi, f) . \tag{8.59}$$

Thus

$$(\psi, f) = \frac{(\psi, g)}{1 - \lambda(\psi, \phi)} , \tag{8.60}$$

and

$$f(x) = g(x) + \lambda \frac{(\psi, g)}{1 - \lambda(\psi, \phi)} \phi(x) \tag{8.61}$$

for all values of λ, *except* $\lambda = 1/(\psi, \phi)$. Thus right at the boundary of the domain of validity of the Neumann series, we seem to run into ambiguities. The reason for this may be most easily seen by considering the case where $g(x) = 0$. Then our equation becomes simply

$$f(x) = \int k_s(x, y) f(y)\, dy = \lambda\phi(x)(\psi, f) , \tag{8.62}$$

which is an eigenvalue problem of the usual type. Equation (8.62) implies that

$$(\psi, f) = \lambda(\psi, \phi)(\psi, f) ,$$

which has a nonzero solution for (ψ, f) if and only if

$$\lambda = 1/(\psi, \phi) ,$$

in which case (ψ, f) can be any constant. Thus the eigenvalue problem has as its solution

$$f(x) = A\phi(x) ,$$

where A is arbitrary. To fix A, a normalization condition is necessary, just as in the finite-dimensional case. The interesting result is that the condition for the existence of an eigensolution is the same condition under which we saw that a solution of the inhomogeneous equation given by Eq. (8.61) did not exist. But this result is familiar from finite-dimensional theory, where we know that if the homogeneous equation has a solution, then the inhomogeneous term of the inhomogeneous equation must be subject to certain conditions in order that a solution should exist. We see immediately from Eq. (8.61) that we must require $(\psi, g) = 0$ to compensate for the vanishing denominator. Thus in Eq. (8.59) we find that (ψ, f) is arbitrary, so the solution to Eq. (8.54), given by Eq. (8.58), becomes

$$f(x) = g(x) + B\phi(x) ,$$

where B is an arbitrary constant. Substituting this result into Eq. (8.54), we see that it is indeed a solution for any B when $\lambda(\psi, \phi) = 1$, that is, when $\phi(x)$ is an eigenvector of the integral operator K_s whose kernel is $k_s(x, y)$. This looks very much like the finite-dimensional Theorem 4.27; the resemblance is com-